

Toward profitable energy futures trading strategies using reinforcement learning incorporating disagreement and connectedness methods enabled by large language models

Tianxiang Cui^a, Yujian Ye^b, Yiran Li^c, Nanjiang Du^a, Xingke Song^a, Yicheng Zhu^a, Xiaoying Yang^a, Goran Strbac^d

^a School of Computer Science, University of Nottingham Ningbo China, 199 Taikang E Rd, Ningbo, 315104, China

^b School of Electrical Engineering, Southeast University, Sipailou 2, Nanjing, 210096, China

^c Network Security Department, Shanxi Police College, Xibefanglu 799, Taiyuan, 030401, China

^d Department of Electrical and Electronic Engineering, Imperial College London, Exhibition Rd, London, SW7 2AZ, UK

HIGHLIGHTS

- Propose a novel framework to optimize trading strategies in energy futures markets.
- Integrate investor sentiment, disagreement, and macroeconomic factors in framework.
- Use chain-of-reasoning and connectedness techniques to classify investor behavior.
- Demonstrate superior returns and risk management over traditional trading methods.
- Enhance energy futures trading performance amid market decarbonization challenges.

ARTICLE INFO

Keywords:

Energy market
Energy future
Reinforcement learning
Large language models
Investor disagreement
Connectedness analysis

ABSTRACT

The energy market plays a fundamental role in the global economy, shaping energy prices, inflation, and financial stability across nations. As the world transitions toward low-carbon energy solutions, optimizing trading strategies in this complex and dynamic market has become increasingly critical for investors, policymakers, and energy brokers. Traditional data-driven models often struggle to capture the multifaceted and interconnected factors influencing energy markets, such as macroeconomic conditions, investor sentiment, and the accelerating shift toward decarbonization. To address these challenges, a novel framework is proposed that combines reinforcement learning with methods for analyzing disagreement and connectedness, alongside advanced natural language processing techniques, to develop trading strategies for energy markets. The proposed method integrates structured time-series data with unstructured textual data to incorporate diverse factors, including the interplay between economic influences, green energy transitions, and investor sentiment. The proposed framework also employs a chain-of-reasoning technique to classify investor types, distinguishing between sentiment-driven disagreement and cross-disagreement, and utilizes a connectedness-based method to model the interrelationships among market variables, providing a comprehensive understanding of market dynamics. As a showcase, this framework is applied to the West Texas Intermediate crude oil market, demonstrating its ability to outperform traditional price-prediction-based trading strategies. Experimental results highlight that the proposed framework delivers superior investment returns while addressing key limitations of existing models in terms of data integration and flexibility. This study underscores the potential of the proposed framework as a robust and adaptable solution for optimizing trading strategies across the broader energy market, with particular relevance to the global transition toward sustainable energy systems.

* Corresponding author.

E-mail addresses: tianxiang.cui@nottingham.edu.cn (T. Cui), yeyujian@seu.edu.cn (Y. Ye), liyrxm@gmail.com (Y. Li), nanjiang.du@nottingham.edu.cn (N. Du), xingke.song@nottingham.edu.cn (X. Song), biyyz146@nottingham.edu.cn (Y. Zhu), xiaoying.yang@nottingham.edu.cn (X. Yang), g.strbac@imperial.ac.uk (G. Strbac).

<https://doi.org/10.1016/j.egyai.2025.100562>

Received 24 January 2025; Received in revised form 15 April 2025; Accepted 11 July 2025

Available online 23 July 2025

2666-5468/© 2025 The Authors. Published by Elsevier Ltd. This is an open access article under the CC BY-NC license (<http://creativecommons.org/licenses/by-nc/4.0/>).

1. Introduction

The energy market is a cornerstone of the global economy, shaping energy prices, inflation, and financial stability across nations [1,2]. Over the past century, the energy landscape has undergone significant transformations driven by technological advancements, geopolitical developments, and shifts in global economic systems [3]. Today, this market is at the center of a critical transition toward low-carbon energy systems, as nations and industries strive to meet ambitious decarbonization goals. This transition is reshaping the energy industry, introducing new complexities and opportunities for market participants, including investors, policymakers, and energy brokers. These stakeholders must navigate an increasingly dynamic and uncertain environment where traditional energy sources coexist with renewable alternatives.

Despite the growing adoption of renewable energy technologies, fossil fuels such as crude oil remain integral to the global energy mix, with forecasts suggesting their continued relevance for at least the next two decades [4,5]. However, the global push for decarbonization is accelerating the integration of renewable energy sources, the implementation of carbon pricing mechanisms, and the expansion of green finance initiatives [6,7]. These developments introduce intricate interdependencies across the energy market, requiring innovative tools to understand, predict, and manage market dynamics. Energy brokers, in particular, face unique challenges in this evolving landscape, including managing price volatility, analyzing market sentiment, and aligning trading strategies with sustainability goals.

Investor sentiment plays a critical role in shaping energy market trends, particularly in volatile environments. Events such as macroeconomic news, geopolitical tensions, and catastrophic events can significantly influence market sentiment, leading to heightened volatility and uncertainty. Recent advances in natural language processing (NLP), particularly the development of large language models (LLMs) such as OpenAI's ChatGPT, have opened new avenues for extracting insights from unstructured textual data, such as news articles, policy reports, and social media [8]. Financial sentiment analysis (FSA) has become an increasingly important tool for forecasting short-term returns, detecting fraud, and understanding market behavior [9]. Beyond sentiment analysis, investor disagreement—reflecting differences in risk perception, expectations, and investment strategies—provides an additional layer of complexity. Disagreement among investors can reveal discrepancies in market sentiment that aggregate measures may fail to capture, particularly in uncertain environments [10].

In addition to sentiment and disagreement, the energy market is influenced by a variety of structural factors, including supply and demand dynamics, financial market movements, regulatory changes, and macroeconomic indicators [11]. The integration of renewable energy sources, the emergence of unconventional resources such as shale oil and gas, and the standardization of energy pricing mechanisms have further added to the complexity of the market [12,13]. Policymakers, energy brokers, and investors rely on accurate market predictions to develop effective strategies, manage risks, and ensure the stability of the energy system. However, traditional data-driven models often struggle to capture the multifaceted and interconnected nature of these factors, underscoring the need for more robust and adaptable analytical frameworks [14].

To address these challenges, this paper introduces a novel framework, RLDC-LLM (Reinforcement Learning Incorporating Disagreement and Connectedness Methods with Large Language Models), designed to optimize trading strategies across the energy market. First, the proposed framework integrates structured time-series data with unstructured textual data, enabling a comprehensive analysis of market dynamics. This integration captures a wide range of factors, including macroeconomic conditions, investor sentiment, and emerging trends such as green energy transitions. Second, the framework extends beyond traditional sentiment analysis by investigating the impact of investor disagreement on market trends. It distinguishes between basic

disagreement (rooted in sentiment differences) and cross-disagreement (arising from varying investment strategies, such as value-based versus technical investing). Using a chain-of-reasoning prompt technique, the framework classifies investor types, enhancing the LLM's ability to interpret complex market behaviors and investor interactions. Third, the framework incorporates a connectedness-based approach to model the interrelationships among market variables. By weighting various factors in the reinforcement learning states, it reflects the intricate relationships between market dynamics, improving the performance of the trading strategy obtained by the reinforcement learning agent. Finally, the framework is applied to the West Texas Intermediate (WTI) crude oil market as a case study, showcasing its ability to outperform traditional price-prediction-based trading strategies. Experimental results demonstrate that the investment returns achieved by the RLDC-LLM approach significantly exceed those of conventional methods, validating its efficacy in real-world trading scenarios.

The remainder of this paper is organized as follows: Section 2 provides a comprehensive literature review, focusing on the application of LLMs in the financial domain and various models employed for crude oil trading. Section 3 outlines the methodology, detailing the LLM-based disagreement analysis, the connectedness analysis, and the research framework. Section 4 presents the experimental results and their analysis. Finally, Section 5 concludes the paper.

2. Literature review

This section begins by reviewing key literature on the use of LLMs in FSA-assisted trading strategies and then examines factors influencing the crude oil futures market, including macroeconomic variables and trading methodologies.

2.1. LLM-Assisted trading strategies and FSA

The advancement of modern LLMs is based on the advances made in the 2010s, when deep learning approaches began to dominate natural language processing. Preliminary investigations in statistical machine translation (SMT) and Hidden Markov Models (HMMs) laid the groundwork for probabilistic language models. The advent of word embeddings such as Word2Vec transformed the paradigm by allowing models to acquire dense vector representations of words. The representations enabled the acquisition of semantic and syntactic links, leading to significant improvements in language tasks. The development of the Transformer architecture in 2017 was a significant advancement by replacing recurrent models with self-attention processes, hence improving scalability and parallelization. Transformers laid the foundation for later models including BERT, GPT-2 and GPT-3, demonstrating the effectiveness of large-scale pretrained models. Models like GPT-3, comprising 175 billion parameters, signify a significant progression, demonstrating that large language models trained on extensive and diverse datasets can perform multiple tasks with minimal task-specific modifications, including applications in financial domains [15].

The integration of LLMs in finance is propelled by their ability to handle complex financial data and generate significant insights across diverse tasks. As a result, LLMs have increasingly been applied in various financial domains, showcasing their transformative potential. Notably, these models have demonstrated proficiency in tasks such as sentiment analysis and explanation production. Models such as FinGPT and BloombergGPT exemplify the capabilities of LLMs in finance, demonstrating their effectiveness in analyzing financial time-series data and enhancing cross-sequence reasoning [16]. The emergence of LLM-based trading agents, such as Sociodojo [17], illustrates the potential of LLMs to innovate investment techniques and trading strategies. Recent advancements have resulted in the rise of LLM-driven trading agents that prioritize decision-making based on an extensive array of external information, encompassing financial news, reports, and stock market data. Through the analysis and refinement of this data, these agents

can produce actionable signals for asset acquisition or divestment. This section analyzes various categories of LLM-based trading agents, including media-driven, reflection-driven, and RL-driven agents, each offering distinct methodologies for employing LLMs in financial market decision-making.

The media-driven architecture represents a fundamental variant of LLM-based trading, wherein news, macroeconomic updates, and tweets are integrated into the model's prompt environment. Recent research have assessed both proprietary LLMs (e.g., GPT-3.5/4) and open-source models for financial sentiment analysis, proving their efficacy in backtesting basic long-short strategies predicated on sentiment scores. Additionally, these techniques [18] demonstrate the capability of LLMs in facilitating educated trading decisions through sentiment analysis. Recent improvements have developed more complex designs, enabling LLMs to not only summarize news material but also analyze the correlations between news and stock price fluctuations. Fatouros et al. [19] created summary modules for daily news, macroeconomic updates, and stock price trends.

Media-driven agents prioritize the analysis and response to real-time financial news in order to inform trading decisions. Conversely, reflection-driven agents implement a more sophisticated methodology by incorporating historical data into their decision-making process. In LLM-based agents, the architecture of memory and reflection is influenced by cognitive science principles. Human learning involves the following: engaging with the environment, assimilating feedback, constructing memories, and utilizing acquired knowledge to complete tasks. In the same vein, the incorporation of memory and reflection into LLM-based trading systems mitigates issues such as hallucinations and provides a more comprehensive understanding of the market landscape.

In contrast to reflection-driven agents, which concentrate on the synthesis and utilization of accumulated memories to improve decision-making, discussion-driven agents engage in internal dialogs among multiple LLMs to refine reasoning and enhance the validity of outputs. Engaging in discussion among LLMs has demonstrated improvements in critical thinking and factual accuracy. This approach is especially beneficial in financial contexts, where nuanced and reasoned judgments are essential. Xing et al. [20] propose a heterogeneous debating framework wherein LLM agents adopt distinct roles—specifically mood, rhetoric, and dependency agents—and participate in discussion to enhance the classification of news sentiment.

Discussion-driven agents improve decision-making through internal conversations and role-based interactions amongst several LLMs, while RL-driven agents optimize feedback loops. Methods like RLHF (Reinforcement Learning with Human Feedback) and RLAI (Reinforcement Learning with AI Feedback) have been successful in aligning LLM outputs with intended behaviors through consistent, high-quality feedback. However, getting feedback efficiently is difficult. In financial trading, backtesting simulates real-market circumstances to generate feedback at low cost, rewarding RL-driven agents. SEP [21] presents a RL approach with memorizing and reflection modules, enabling trading agents to improve their strategies based on past forecasts. RL is a renowned technique for decision-making in games and trading, optimizing actions over time [21]. LLM-powered models train the system utilizing Proximal Policy Optimization (PPO) using news headline embeddings in stock feature spaces [22]. LLM-generated embeddings and RL training help the agent adapt to complex market conditions and make better decisions.

2.2. Factors affecting crude oil and trading strategies

The impact of crude oil trends on the global economy and society has led to extensive research on the factors influencing these trends. Recent studies demonstrate a significant correlation between the predictability of crude oil trends and the volatility of commodity trends [23]. Research on the predictability of oil trends primarily examines macroeconomic variables, including stock trends [24], the dollar

exchange rate, monetary policy, U.S. and global macroeconomic aggregates, and various oil shocks. Additional factors including supply and demand dynamics [25], geopolitical considerations [26,27], economic uncertainty [28], and investor attention [29] have been analyzed.

Combining diverse economic market data provides another promising approach to crude oil trading strategies. Beyond trade and position data, factors such as other energy sources (both fossil and renewable), major commodities like gold and copper, stock index, bond indices, and the US dollar index play critical roles. Coal, for instance, accounts for nearly 25% of the global energy supply and generated over two-fifths of electricity in 2013 [30]. The interaction between crude oil and coal is particularly noteworthy: as coal price rise, crude oil usage increases, and vice versa [31]. Natural gas is also a crucial energy commodity that meets industrial and residential needs [32,33]. The IEA indicates that natural gas constitutes under 25% of the global energy supply, with its markets intricately linked to crude oil regarding both demand and supply [32].

Turning to the broader implications of energy transitions, research suggests that existing energy sectors may transition to sustainability while fostering innovation in renewable energy [34]. In countries with substantial fossil fuel industries, the shift to renewable energy presents a chance to use existing competencies, technology, and expertise. The oil and gas sector in Norway has enabled the creation of an innovation system focused on offshore wind energy generation [35]. This underscores the importance of analyzing renewable energy in relation to energy transitions. Furthermore, Li et al. [36] demonstrate that Chinese green energy stocks significantly influence fluctuations in the crude oil market, whereas the natural gas stock market exerts an even greater impact. The correlation between crude oil returns and clean energy firm returns stays continuously strong, irrespective of temporal fluctuations. Additionally, an asymmetry is present in the informational interconnectedness between oil and clean energy firms, whereby bad news has a more pronounced impact than positive news [37].

Moreover, evidence suggests a long-term equilibrium between the gold and crude oil markets, characterized by a bidirectional causal relationship between gold and oil returns [38]. Key commodities like gold, copper, and crude oil are positively correlated and serve as effective hedges against inflation [39,40]. Utilizing these commodities states as variables can substantially improve crude oil trading strategies. In addition to commodity markets, the relationship between stock and crude oil is complex [41]. The presence of nonlinear relationships among these markets complicates the analysis, as traditional linear causality tests may not accurately reflect the actual connections [42]. Fluctuations in the crude oil market influence stock markets and are, in turn, influenced by them, highlighting the necessity for integrated analytical frameworks. The relationship between crude oil and stocks is multifaceted and varied. Stock values of oil-producing companies typically demonstrate a positive correlation with crude oil prices. Conversely, firms in other sectors frequently face adverse effects, as escalating oil prices result in higher operational expenses. This study examines the relationship between oil and stock markets to address how fluctuations in stock market affect crude oil performance and vice versa, by integrating essential market indices, such as the Dow Jones Industrial Average, and NASDAQ Composite Index, into the analytical framework. Moreover, previous studies consistently highlight an inverse relationship between the US dollar index and crude oil prices, with oil prices typically rising as the US dollar weakens [43].

The correlation between stock market volatility and oil market trends has become a prominent area of study. Studies demonstrate that the integration of oil market data improves the accuracy of predictions related to stock market trend volatility [44]. The predictive relationship between the stock market and oil trend volatility is variable, as noted in the review by [45]. For example, [46] demonstrated that stock trends did not offer a distinct advantage over other factors in predicting oil trend volatility, whereas additional studies indicate that stock market data can enhance forecasts of oil trend volatility [47]. The stock market

uncertainty index contributes additional information for forecasting the volatility of crude oil futures [48]. The forecasting capacity for oil markets is enhanced by stock market information, as evidenced by research on volatility spillovers [49]. This research emphasizes the time-varying comovements or causal relationships between the volatilities of oil and stock market trends [50].

Shifting the focus to another critical macroeconomic variable, bond yield, encompassing both government and corporate bonds, is a fundamental topic within macroeconomic variables. The United States possesses the largest and most influential bond market globally. U.S. Treasury securities are recognized as a global safe haven, primarily due to their minimal default risk and the U.S. dollar's status as a reserve currency. The yields on short- and long-term Treasuries provide significant insights into the present and anticipated conditions of the economy and inflation [51]. Numerous studies have examined the relationship between bond yield and crude oil due to their significance. Kang et al. [52] demonstrate that a positive oil market-specific demand shock leads to reductions in real bond returns. Morrison [53] identifies a statistically significant impact of oil prices on the total sovereign bond returns of both oil-exporting and oil-importing nations. Sovereign governments' bond portfolios are sensitive to fluctuations in investor risk perception, independent of their oil trade status. Additionally, Dai and Kang [54] demonstrate that bond yields can effectively predict oil prices both in-sample and out-of-sample.

The primary focus of crude oil trading strategies is on price prediction as a critical contributor to decision-making. Econometric and AI-based models have been extensively employed to forecast oil prices. In order to forecast price fluctuations using time-series data, econometric models such as ARIMA (Autoregressive Integrated Moving Average) [55], QRF (Quantile Regression Forests) [56], GARCH (Generalized Autoregressive Conditional Heteroscedasticity) [46], and VAR (Vector Autoregression) [57] are implemented. In contrast, AI models, such as decision tree (DT) methods [58], neural networks (NN) [59], and long short-term memory (LSTM) networks [60], are designed to capture intricate nonlinearities in price behavior. These models depend significantly on historical price data and frequently forecast future price levels, based on the premise that past price movements reflect future trends.

Nevertheless, trading strategies that rely on price prediction are subject to substantial constraints. The intrinsic volatility and complexity of the oil market present significant challenges for price prediction models, which impedes the accuracy of forecasting endeavors. Additionally, these models may experience overfitting and interpretability issues, which can result in unreliable predictions in specific market conditions [61]. The proposed trading strategy is a departure from price prediction, with an emphasis on the optimization of buy and sell decisions in accordance with market signals.

2.3. LLM on time series data

Recent advancements have demonstrated the surprising proficiency of LLMs in time series forecasting, especially when paired with innovative encoding strategies. For example, Gruver et al. [62] achieved a 12.7% sMAPE reduction over ARIMA on M4 datasets by representing numerical sequences as tokenized digits, leveraging Transformers' self-attention to capture non-linear patterns. In clinical settings, hybrid models that combine LLM-processed medical text with physiological signals have improved sepsis prediction AUC to 0.89, a 0.06 increase compared to LSTM baselines, using MIMIC-IV data [63]. Similarly, the iTransformer architecture introduced variate tokenization and inverted traditional attention mechanisms to better model cross-variable dependencies, yielding 12%–18% accuracy gains in power load forecasting [64]. A series of innovative approaches leveraging LLMs have emerged in the realm of time series forecasting. Notably, CALF [65] presents a cross-modal fine-tuning framework designed to align the distributions of textual and temporal data within LLMs. This framework

effectively addresses modality gaps through feature regularization and output consistency losses, achieving state-of-the-art performance and enhancing few-shot and zero-shot capabilities. ST-LLM [66] contributes significantly by proposing a spatial-temporal LLM for traffic prediction. By tokenizing timesteps and locations, integrating spatial-temporal embeddings with fusion convolutions, and adopting a partially frozen attention strategy, it captures global dependencies and demonstrates robust performance in few-shot and zero-shot scenarios. Additionally, One Fits All [67] leverages frozen pretrained language or vision models for general time series tasks, revealing that unmodified self-attention layers can bridge domain gaps similarly to PCA, yielding competitive results across forecasting, classification, and anomaly detection. Moreover, Gruver et al. [62] show that LLMs such as GPT-3 can serve as excellent zero-shot time series forecasters by tokenizing numerical series into text. However, performance varies depending on model size and tokenization design, as evidenced by the limitations of GPT-4 due to number encoding. Collectively, these works illustrate the versatility of LLMs in time series analysis through cross-modal alignment, spatial-temporal tokenization, pretrained knowledge transfer, and flexible tokenization strategies.

Building on these successes, subsequent research has further advanced LLM applications in time series analysis. For instance, the TIME-LLM framework [68] reprograms pretrained LLMs by aligning time series data with text prototypes and using Prompt-as-Prefix (PaP) to enhance reasoning, achieving state-of-the-art results while keeping LLM backbones frozen. AutoTimes, introduced by Liu et al. [69], revitalizes the autoregressive capabilities of LLMs by tokenizing time series and incorporating textual timestamps for chronological alignment, achieving competitive accuracy with minimal parameters and high computational efficiency. Yu et al. [70] expand LLM application to explainable temporal financial forecasting, demonstrating that LLMs can reason over multi-modal data such as news and prices and generate interpretable outputs, even using public models like Open LLaMA. Although there are performance gaps compared to proprietary LLMs like GPT-4, this work broadens the application field of LLMs. Similarly, Wang et al. [71] introduce LLM-based agents to iteratively filter and reason over unstructured news events, aligning textual context with time series fluctuations to refine forecasts iteratively.

However, not all research is optimistic about the role of LLMs in time series forecasting. Tan et al. [72] critically evaluate the utility of LLMs, demonstrating through ablation studies that LLM components often add negligible value. In fact, replacing them with simpler attention layers or removing them entirely can improve performance, challenging the claims of LLM superiority in dealing with time series data and calling for a more rational exploration of LLM applications on time series data.

3. Methodology

The proposed Reinforcement Learning incorporating Disagreement and Connectedness methods with Large Language Models (RLDC-LLM) is designed for WTI Crude Oil Trading. As shown in Fig. 1, a LLM-based feature extractor to infer both the time-series data and the financial messages from the social media. In particular, time-series information is extracted from time-series data of assets, containing the open, closed, highest, lowest value, and trading volume. In addition, the adjusted sentiment, basic disagreement, and cross-disagreement characteristics are extracted from sentiment LLMs and CIC LLMs.

As shown in Fig. 7, the extracted features for all assets will concatenated after calculating the connectedness between each asset and WTI crude oil, then the multi-head attention module and the LSTM Layer are used to capture the critical features from the observation features. The actor in Proximal Policy Optimization (PPO) module will give the certain trading action, i.e., “sell”, “hold”, and “buy”.

To address the inherent challenges in evaluating LLM outputs and ensuring their reliability when integrated with LSTM networks, we

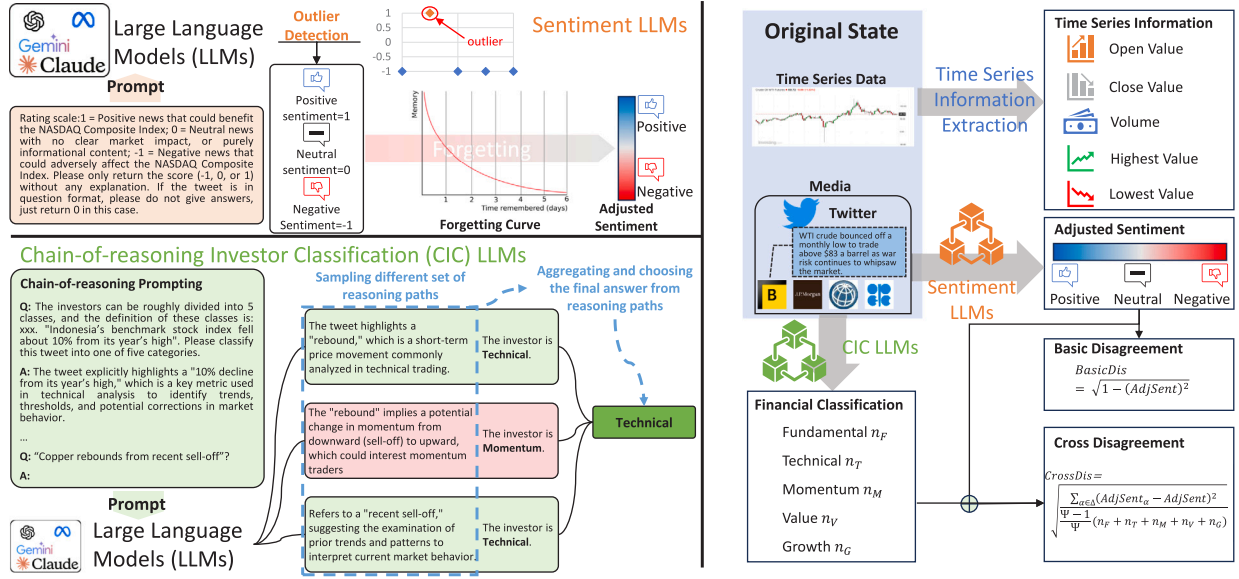


Fig. 1. A brief framework of feature extractor. The time series information is extracted from the time-series data, including the open, close, highest, and lowest values of the different objectives, and the trade volume. To infer the sentiment from different financial media, the sentiment LLMs are introduced to classify the twitter message into three main classes, i.e., positive, neutral, and negative. Besides, the outlier detection is designed to reduce the abnormal prediction from LLMs, and the forgetting mechanism is to reduce the sentiment affect depending on the time passed. The basic disagreement of different investors are calculated based on the adjusted sentiment. The investor classification is calculated by the Chain-of-reasoning Investor Classification (CIC) LLMs, which adapted from a chain-of-reasoning and aggregating methods. The cross disagreement between different types of investors is calculated by utilizing both adjusted sentiments from sentiment LLMs and investor classification from CIC LLMs.

adopted a multi-layered validation strategy. First, we implemented chain-of-thought prompting to elicit structured reasoning processes from the LLM, enhancing output interpretability and reducing logical inconsistencies. This technique encourages the model to decompose complex tasks into intermediate steps, thereby improving the factual accuracy of its responses. Second, we incorporated outlier detection mechanisms at the data preprocessing stage. By leveraging statistical methods and contextual analysis, we systematically identified and filtered anomalous outputs that deviated significantly from expected patterns. This dual-pronged approach, combining quantitative metrics with qualitative assessments, ensured the integrity of the dataset before its integration into the LSTM architecture. Finally, we embedded the LLM within a RL framework, where its outputs serve as dynamic inputs to the RL state space. The RL paradigm's inherent fault tolerance enables it to adaptively weigh the confidence levels of LLM-generated features, mitigating the impact of imperfect data on downstream decision-making. This symbiotic integration allows the system to maintain robust performance even when faced with uncertainties in LLM predictions.

3.1. Sentiment analysis

Investor sentiment analysis of the market is a critical factor in trading strategies [73]. In Fig. 1, the proposed Sentiment LLMs framework aims to classify financial content in social media, such as Twitter messages, into three sentiment categories: positive, neutral, and negative, while incorporating mechanisms to improve robustness and relevance over time. The main components of the proposed sentiment LLMs framework can be summarized in three folds:

3.1.1. LLM prompt design for sentiment classification

The Large Language Models (LLMs), such as GPT-4o, Claude 3.5, and Meta Llama 3.1 8B, are tasked with sentiment classification for social media finance messages based on a well defined rating rule:

Positive sentiment (+1): Messages that convey optimism about the current financial index or market conditions suggest a positive market

impact. These messages often indicate that investors may be inclined to buy stocks or assets due to the favorable sentiment.

Neutral sentiment (0): Messages that convey a neutral or unclear stance regarding the market or financial index. These messages may lack explicit positivity or negativity, leading investors to maintain their current positions (i.e., hold their stocks or assets).

Negative sentiment (-1): Messages that express pessimism about the financial index or market performance, indicating a potential negative market impact. Such sentiment may encourage investors to sell their stocks or assets to mitigate potential losses.

The prompt explicitly instructs the LLMs to classify each message into one of these three sentiment categories by assigning a score of +1 (positive), 0 (neutral), or -1 (negative). For ambiguous cases, such as tweets framed as questions or lacking sufficient context, the LLMs are directed to assign a neutral score of 0. To enhance the classification process, the LLMs are provided with a set of example messages labeled with corresponding sentiments. This ensures better alignment with the rating rules, especially in handling financial terminology and context. The LLMs are then applied to classify social media messages from financial institutions and market participants into the three sentiment classes: positive, neutral, and negative.

3.1.2. Outlier detection

To enhance the reliability of the sentiment classification, an outlier detection mechanism is incorporated. This mechanism identifies and mitigates abnormal predictions that deviate significantly from expected sentiment trends. If a sentiment score is anomalously far from the surrounding pattern (e.g., a sudden positive sentiment prediction between consistent negative predictions), it is flagged as an outlier and excluded from further analysis.

3.1.3. Forgetting mechanism

A forgetting mechanism is introduced to account for the diminishing relevance of sentiment over time. This mechanism applies a decay function for sentiment estimated by the "Forgetting Curve" to reduce the influence of sentiment evaluation for older messages. In detail, the sentiment effect diminishes exponentially with time, ensuring that

more recent sentiment signals carry greater weight in the adjusted sentiment index. This mechanism accounts for the observation that investor sentiment is not discrete or confined to daily intervals; instead, it exhibits a contagion effect that persists for approximately seven days. Mathematically, it can be expressed as:

$$\mathfrak{F} = e^{-\frac{t}{\mathfrak{D}}} \quad (1)$$

where $\mathfrak{F}$ represents the retention rate, or the proportion of information remembered after time t , e is the base of the natural logarithm (approximately 2.718), t is the time in days, and $\mathfrak{D}$ is a constant decay rate that signifies the relative strength of memory, typically set to 1.84, though it may vary depending on context. As we can see, when $t = 7$, the remembered proportion $\mathfrak{F} = 0.022$, and when $t = 8$, $\mathfrak{F} = 0.013$. For simplicity in subsequent computations, we assume the remembered sentiment on day 8 is negligible and set it to neutral (0).

The final output of the Sentiment LLMs is an adjusted sentiment feature, which is represented on a scale ranging from negative (−1) to positive (+1) investment sentiments for markets, providing a refined measure of different investors' sentiments toward financial social media messages.

Our model employs a single exponential decay rate ($\mathfrak{D}$) for sentiment across all timeframes, which serves to produce a cleaner and more stable signal. This approach contrasts with allowing different decay rates per timeframe, which introduces extra flexibility but may inadvertently capture random noise. This is a classic trade-off: while more parameters (different forgetting rates) can potentially overfit idiosyncratic fluctuations, a single decay rate smooths out short-term jitters and emphasizes broader sentiment movements. Research in time-series forecasting supports this notion, indicating that adaptive or highly flexible weighting schemes often yield unstable forecasts by overreacting to random fluctuations. Moreover, a uniform decay rate aligns well with how market participants collectively process news over time. Human attention and memory often follow an exponential forgetting curve, where the impact of information decays roughly exponentially in the absence of reinforcement. In behavioral finance, investors are influenced by recency bias (overweighting recent news) and anchoring (sticking to initial impressions), which together imply a gradual decay of a news event's influence rather than a very short or extremely long memory. By using one decay rate for all timeframes, the model captures this typical investor "memory span" for news. Empirical studies further support the notion that news impacts have a measurable half-life. For instance, it has been shown that news-based order flow can be decomposed into a fast-decaying component (with a half-life of 1 day) and a slower-decaying component (with a half-life of approximately 25 days). The fast, high-frequency sentiment impact tended to reverse (exhibiting a negative predictive effect), whereas the slower sentiment impact persisted and positively predicted returns. This suggests that the lasting part of sentiment is more useful for prediction, whereas overly short-lived sentiment may inject noise or transient effects. A single decay rate effectively emphasizes the average persistence of sentiment that investors carry, filtering out ultra-short-term reactions that might mean-revert or confuse the signal.

3.2. Disagreement

Disagreement-prone assets are generally overvalued. Because of investor disagreement, especially after earnings presentations, estimates are overstated. Assets with stronger arbitrage constraints experience a greater mispricing effect. Thus, considerable disagreement can affect trade volume and mispricing, especially when investor expectations dictate market activity during announcements. Therefore, incorporating investor disagreement is essential for the development of effective crude oil futures trading strategies. Accurate measurement and quantification of disagreement require the enhancement of sentiment analysis via a forgetting mechanism, as outlined in Eq. (1). The forgetting mechanism modifies sentiment by integrating both present

and historical sentiment scores, with the weights diminishing over time in accordance with the forgetting curve. The sentiment on day t incorporates contributions from prior days' sentiment values, with each previous sentiment weighted by a retention rate $\mathfrak{F}$, which indicates the retention of sentiment over time. This process quantifies the contagion effect of sentiment over a seven-day period. The modified sentiment ϖ_{it} is represented as, $\varpi_{it} = \varrho_{it} + \sum_{j=1}^7 \mathfrak{F}_j \cdot \varrho_{i(t-j)}$, where ϱ_{it} represents the sentiment score for subject i on day t , $\mathfrak{F}_j = e^{-\frac{j}{\Gamma}}$ is the retention rate for sentiment from j days prior, with Γ denoting the memory strength parameter. The sentiment score $\varrho_{i(t-j)}$ refers to the sentiment for subject i on day $t - j$, and the sentiment values range from −1 (indicating complete bearish sentiment) to +1 (indicating complete bullish sentiment). The summation extends over the previous seven days, i.e., $j = 1, 2, \dots, 7$. For example, on day 1, the sentiment is solely determined by ϱ_{it} , with no contribution from prior days. On day 2, the adjusted sentiment includes the weighted contribution of day 1's sentiment, adjusted by $\mathfrak{F}_1$. By incorporating this seven-day retention process, the adjusted sentiment accounts for the diminishing influence of past sentiments, providing a more nuanced and time-sensitive measure of investor sentiment.

To measure investor disagreement, we calculate the standard deviation of sentiment expressed across messages for a particular asset on a given day, as in [74]. At the asset-day level, denoted as D_{it} , we aggregate sentiment by averaging the sentiment across all tweets about asset i between 4:00 PM on day $t - 1$ and 4:00 PM on day t . As the sentiment is binary (with −1 representing bearish and +1 representing bullish), the variance in sentiment for asset i on day t is computed as $1 - \varpi_{it}^2$. Consequently, the disagreement measure, defined as the standard deviation of sentiment, is given by:

$$D_{it}^b = \sqrt{1 - \varpi_{it}^2} \quad (2)$$

The basic disagreement measure, D_{it}^b , is calculated as the standard deviation of sentiment (essentially a measure of variability or disagreement in the sentiments expressed by different investors or sources), used for non-zero asset-day observations. Since the sentiment score ϖ_{it} is a normalized value between −1 and +1, squaring it results in a value between 0 and 1. Subtracting this squared value from 1 and taking the square root produces a measure of how much disagreement exists about the asset. In cases where there are no messages for an asset on a given day, the standard deviation of sentiment cannot be computed. In such situations, we assume that the lack of messages indicates no immediate trading interest, so we normalize the disagreement to 0. This approach aligns with the concept of latent agreement, where minimal disagreement is associated with minimal trading activity, as suggested in [75].

Inspired by [75], we construct two types of cross disagreement: cross-investor group disagreement and cross-market disagreement, to capture different dimensions of investor sentiment and its impact on trading strategies. Cross-investor group disagreement measures the variation in sentiment between different groups of investors who follow distinct investment philosophies, such as Technical, Fundamental, Value, Growth, and Momentum. The disagreement within each group is first computed using the formula in Eq. (2). Then, the weighted average of the within-group disagreement is calculated, where the weights correspond to the number of investors in each group. The reasoning behind this approach is that larger investor groups have a higher potential to influence market behavior, and therefore their sentiment divergence should have a more significant impact on market decisions. The formulation is represented as:

$$D_{i,at}^c = \sqrt{\frac{\sum_{\alpha \in \hat{A}} n_{\alpha} (\varpi_{at} - \varpi_i)^2}{\frac{\Psi-1}{\Psi} (n_F + n_T + n_M + n_V + n_G)}}, \quad (3)$$

where $\hat{A} = \{\text{Fundamental, Technical, Momentum, Value, or Growth}\}$, n_{α} is the number of individuals in group $\alpha \in \hat{A}$, ϖ_{at} is the adjusted

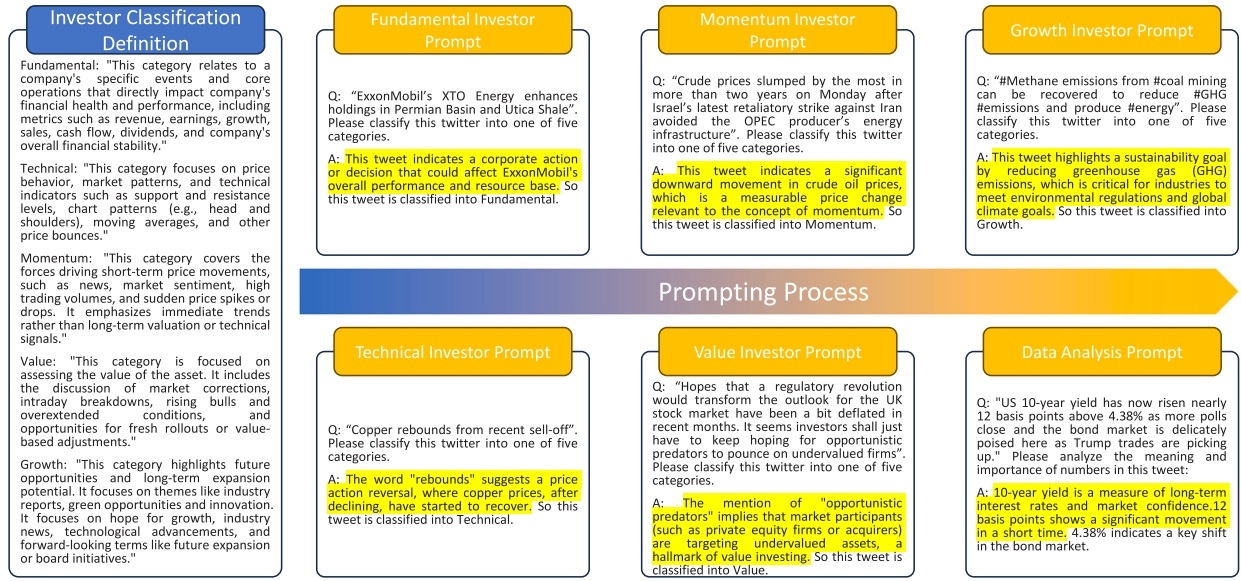


Fig. 2. Chain-of-reasoning prompting and examples. To improve the CIC LLMs' performance, we provide the definition of the five different investment strategies, and the samples of different investors' classification results as prompt.

sentiment of group α from 4:00 pm (close) on date $t - 1$ to 4:00 pm on date t , and Ψ is the number of investment philosophies.

Cross-market disagreement measure captures the sentiment divergence between different market segments, such as crude oil, natural gas, bonds, and other financial assets. The sentiment is adjusted and aggregated across these markets, and the standard deviation is calculated to quantify the level of disagreement across them. Cross-market disagreement provides insight into how market-wide differences in sentiment—stemming from diverse asset classes—can influence the behavior of investors and traders in the crude oil market. The formulation is represented as:

$$D_{i,\beta t}^c = \sqrt{\frac{\sum_{\beta \in B} n_{\beta} (\varpi_{\beta t} - \varpi_t)^2}{\sum_{\beta \in B} (n_{\beta})}}, \quad (4)$$

where B represents the set of discussed assets, n_{β} is the number of individuals discussing assets β , $\varpi_{\beta t}$ is the adjusted sentiment for an asset β from 4:00 pm on date $t - 1$ to 4:00 pm on date t , and ϖ_t is the average sentiment across all assets for the same period.

Noted that, in the following discussion, both types of disagreement will be referred to as "cross disagreement" because, in the cross-market state case, we incorporate both types. The only exception to this is the crude oil case, where we only consider cross-investor group disagreement.

3.3. Chain-of-Reasoning prompting

Although sentiment analysis and disagreement measures can quantify investor sentiment and the level of divergence, they do not inherently provide insights into the underlying investment logic driving those sentiments. In particular, simply knowing whether investors are bullish or bearish does not reveal whether they are adopting a value, growth, technical, or momentum investing approach. This gap complicates the classification of investors into specific groups and subsequently hinders the accurate calculation of cross-investor group disagreement.

To address this challenge, we propose the use of a chain-of-reasoning prompting framework, inspired by [16], to simulate the cognitive processes of investors more explicitly. By breaking down the reasoning process into intermediate steps, the chain-of-reasoning approach allows us to infer not only the sentiment of an investor but

also their underlying investment philosophy. This decomposition makes it possible to map each investor's sentiment to a specific investment strategy, facilitating more accurate classification and, in turn, more precise calculation of cross-investor group disagreement. Incorporating this reasoning step into the disagreement model enhances the interpretability and accuracy of investor behavior analysis, enabling us to better understand how different investment philosophies contribute to market dynamics.

The prompting mechanism for CIC LLMs is shown in Fig. 2. First, we define five investment strategies: Technical, Fundamental, Momentum, Value, and Growth to improve the model's investor classification. Foundational definitions help the model analyze and classify investor behavior. Investor classification samples based on sentiment analysis and reasoning procedures that mirror real-world investment decision-making are given to the LLM. These predetermined techniques assist the model to perform a structured reasoning sequence, simplifying investor classification. The reasoning processes let the algorithm identify market patterns, assess sentiment, and classify investors by investment philosophy. This technique ensures that the algorithm examines all investor behavior before classifying them into one of the five strategies, improving prediction accuracy.

Fig. 3 presents an example of the answers generated by the LLM after utilizing the Chain-of-Reasoning (CoR) prompting technique. The approach breaks complex problems into smaller phases, allowing the model to focus on each reasoning step, thus improving clarity and accuracy in decision-making. By systematically addressing each component of the problem, the model ensures that all relevant factors are considered, leading to more thoughtful and reasoned conclusions. For instance, considering the statement in Fig. 3 (Example 1):

The sentence in question: "Crude prices slumped by the most in more than two years".

Chain of Reasoning: "Crude prices slumped" identifies a substantial downward movement in oil prices, representing a measurable change. "By the most in more than two years" emphasizes the magnitude and rarity of the decline, requiring an additional step to interpret its significance. This decomposition helps the model focus on each reasoning step before concluding that the classification involves Momentum.

CoR provides a flexible and efficient alternative to fine-tuning, offering several distinct advantages that make it particularly suitable for applications like investor classification. One of the primary benefits of CoR is its ability to enhance model adaptability without altering

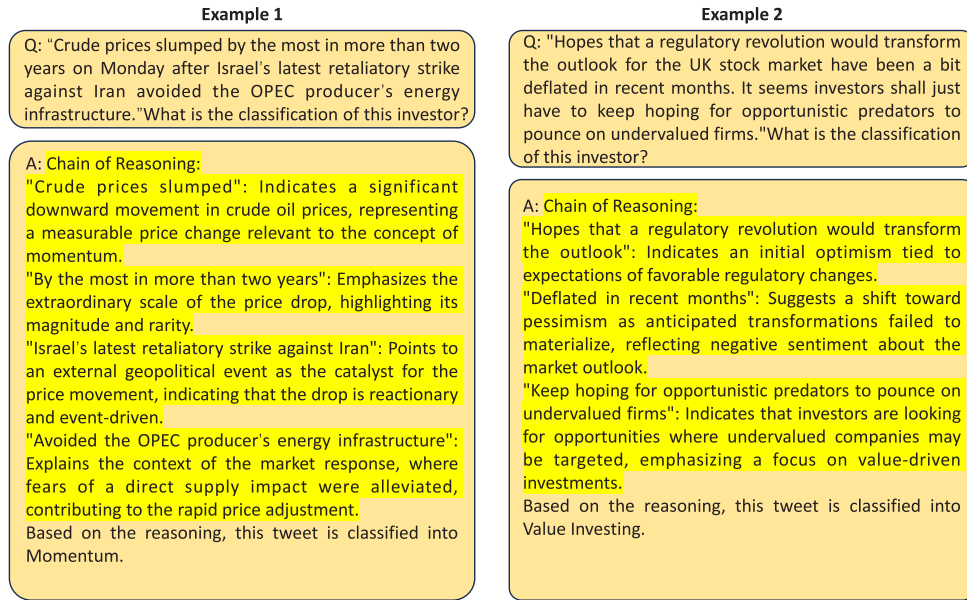


Fig. 3. An example of LLM answers after utilizing the chain-of-reasoning prompting. The chain-of-reasoning answers decompose the multi-step problems into a few intermediate steps, and provides interpretability to help understanding.

the underlying parameters. This means that CoR can be implemented quickly and cost-effectively, as it does not require the extensive computational resources or time associated with fine-tuning. By guiding the model through a structured reasoning process, CoR allows it to leverage its existing capabilities to break down complex tasks into manageable steps, facilitating more accurate and nuanced interpretations of investor behavior. In contrast, traditional fine-tuning procedure involves adjusting the internal weights of a LLM based on task-specific data, which can be extremely resource-intensive and time-consuming. This process requires access to large, high-quality labeled datasets, which may not always be available. Additionally, fine-tuning carries the risk of overfitting, where the model becomes too tailored to the training data and loses its ability to generalize to new, unseen scenarios. This can lead to decreased performance when the model encounters data that differs from the training set, limiting its applicability and robustness in dynamic environments. Moreover, CoR enhances the interpretability of model outputs by explicitly simulating the cognitive processes that underlie investment decisions. By breaking down reasoning into intermediate steps, CoR provides insights not only into the sentiment of investors but also into their underlying investment philosophies, such as whether they are adopting value, growth, technical, or momentum strategies. This decomposition enables a more precise classification of investors into specific groups, facilitating accurate calculations of cross-investor group disagreement and offering deeper insights into market dynamics.

3.4. Connectedness methods

The idea of connectivity is crucial for comprehending complex market linkages, especially in complex financial fields like crude oil trading [76]. Connectivity boosts trading strategy prediction, according to our research. We can better weight and integrate numerous data sources using a connectedness-based method, guaranteeing that the model includes market sentiment, macroeconomic indicators, and investment patterns. This holistic approach helps the model capture the intricate interactions between geopolitical events, economic data releases, and investor mood in the crude oil market. Financial market connectivity is important, according to recent studies. Diebold and Yilmaz [76] highlighted the importance of interconnectivity in financial market volatility, emphasizing the importance of cross-market spillovers for accurate predictions in interdependent markets like crude

oil futures. Our model's connectedness improves its capacity to discover and exploit these correlations, resulting in more effective trading strategies.

In this section, we introduce two key concepts of connectedness, static connectedness and dynamic connectedness, which are integral to our subsequent experiments. This will help contextualize our experimental design by improving knowledge of these ideas. Generally, static connectivity reveals asset or market co-movements and interdependencies through long-term, stable linkages. This connectivity helps us capture long-term relationships. Dynamic connectivity, on the other hand, shows how these ties change with market conditions and external influences. Our model incorporates both static and dynamic connectedness, which enhances its predictive power and the robustness of its trading strategies.

As mentioned previously, the data used in this study include WTI crude oil futures, coal, gold, copper, NASDAQ index, Dow Jones index, natural gas, wind energy index, solar energy index, US Dollar index, CBOT2 (Chicago Board of Trade 2-Year Treasury Note Futures), and CBOT10 (Chicago Board of Trade 10-Year Treasury Note Futures). These data were sourced from the Wind database, with the sample period spanning from December 31, 2015, to November 15, 2024, encompassing 2,247 daily observations per market.

To examine the return spillover effects, we computed the log returns using the formula $\zeta_t = (\ln(\zeta_t) - \ln(\zeta_{t-1})) \times 100$, where ζ_t and ζ_{t-1} represent the price and log return, respectively. The descriptive statistics for all return series are presented in Table 1. Most assets show negative skewness (e.g., crude oil at -2.141), indicating that the return distributions are skewed to the left, meaning more frequent negative returns. Many assets exhibit high kurtosis, particularly crude oil (43.564), suggesting heavy tails or a high frequency of extreme values (outliers). This is consistent with assets that are more volatile and subject to sudden market shocks. The Jarque-Bera (JB) test statistics are very high for most assets, with crude oil having the largest value (218748.127), indicating that the return distributions are far from normal. The large JB statistics, combined with high kurtosis and skewness, suggest that the return series of these assets exhibit significant deviations from normality. The ADF (Augmented Dickey-Fuller) test statistics show that all the return series are stationary, as indicated by negative values for the ADF statistics (e.g., crude oil: -9.300). This suggests no unit roots, implying that the series are mean-reverting. $Q(n)$ represents the Ljung-Box statistic for up to the n th order serial correlation. $Q(5)$ and

Table 1
Descriptive statistics of return series.

	Mean	Standard deviation	Skewness	Kurtosis	JB	ADF	Q(5)	Q(20)
Crude oil	-0.013	2.789	-2.141	43.564	218748.127	-9.300	18.844	85.468
Solar	0.006	2.268	0.036	6.706	4076.101	-10.325	10.174	52.442
Natural gas	0.007	3.797	0.631	9.399	8337.448	-49.524	11.190	34.093
Coal	0.059	3.779	0.001	15.449	17754.376	-8.383	9.666	39.854
Wind	0.001	1.868	-0.281	3.709	1255.723	-13.040	7.985	20.295
Gold	0.039	0.915	-0.057	3.616	1216.559	-34.283	6.065	27.763
Copper	0.030	1.381	-0.113	1.528	221.262	-48.841	2.489	17.242
Dow Jones	0.042	1.127	-1.008	23.907	53283.324	-14.724	104.739	397.218
NASDAQ	0.061	1.379	-0.656	8.154	6311.329	-15.439	48.127	208.929
2 y bond	-0.003	0.181	0.050	12.390	14238.728	-28.108	128.383	155.975
10 y bond	-0.006	0.413	0.125	1.825	314.521	-37.738	52.090	72.712
US dollar index	0.003	0.416	-0.021	2.071	398.342	-20.467	3.075	23.093

Q(20) tests check for autocorrelation up to 5 and 20 lags, respectively. Most assets (except for natural gas) show significant autocorrelation, particularly in the Dow Jones (Q(5) = 104.739, Q(20) = 397.218) and NASDAQ series.

For Table 1, we acknowledge that the asset returns exhibit significant non-Gaussian characteristics, such as high skewness and kurtosis, which are consistent with well-documented features of financial markets. For instance, crude oil returns display negative skewness (-2.141) and extreme kurtosis (43.564), indicating a higher frequency of extreme negative returns compared to a normal distribution. Similarly, other assets like natural gas and the Dow Jones index also show substantial departures from normality. These non-normal distributions motivate our choice of flexible and robust models, such as the RLDC-LLM framework, which can capture complex nonlinear patterns and heavy-tailed behavior. To appropriately evaluate performance under such conditions, we employ risk-sensitive metrics including the Sharpe ratio and conditional value-at-risk (CVaR), which are more informative than traditional measures under skewed and fat-tailed distributions. While the large sample size ($n \approx 2247$ daily returns per asset) suggests that estimators may approach normality due to the Central Limit Theorem, our modeling approach avoids assuming Gaussianity in the raw data. Instead, we utilize nonparametric and model-free methods that do not rely on normality assumptions. The observed skewness reflects inherent asymmetries in market dynamics rather than modeling bias, as our framework learns from these patterns without imposing symmetry constraints. To address potential sensitivities to outliers and volatility clustering in future applications, we have incorporated several robustness measures: GJR-GARCH modeling for volatility estimation, outlier detection in sentiment analysis, and a forgetting curve to mitigate the impact of past shocks. These mechanisms ensure the model's reliability even when confronted with non-Gaussian data.

Then, to explore the relationships between the assets, we calculated the unconditional correlations among the 12 variables, with the results shown in Table 2. The correlation matrix reveals that the 12 variables are highly correlated with each other. Specifically, the correlation between crude oil futures and the Dow Jones stock market is notably high, indicating a strong relationship between these two markets. This suggests that fluctuations in the stock market are likely to have a significant impact on crude oil futures, highlighting the interconnectedness between these markets.

For Table 2, crude oil exhibits mostly weak correlations (less than 30%) with traditional assets like equities and bonds, consistent with empirical finance literature. This aligns with findings that oil's price movements are historically independent of financial assets outside of extraordinary periods [77]. Oil's higher correlation with copper and the Dow Jones is plausible, as both oil and copper are pro-cyclical commodities driven by global industrial demand [78]. Oil can also moderately correlate with equity indices during specific macroeconomic regimes, such as expansions or recessions impacting corporate earnings and commodity demand [79]. The 10-year bond's strong negative correlation with gold reflects their roles as safe-haven assets that often move inversely during market stress [80]. Bonds typically have low

or negative correlations with riskier assets outside of risk-off episodes, consistent with their counter-cyclical behavior. Weak cross-asset correlations (less than 30%) are beneficial for diversification, as uncorrelated assets reduce portfolio volatility [81]. The RLDC-LLM model thrives in this environment by leveraging dynamic connectedness (capturing time-varying co-movements) and disagreement metrics (flagging shifts in market expectations) to enhance strategy robustness [82]. Extremely high correlations (greater than 85%) would imply near-collinearity, reducing diversification benefits and rendering multi-asset strategies less effective [83]. Such correlations are rare in financial markets, as asset classes respond to distinct fundamentals. The observed low-to-moderate correlations are credible and align with realistic market dynamics, validating the RLDC-LLM approach.

To examine the volatility spillover effects, it is essential to first estimate the volatilities of the assets. In line with the methodology proposed by BenSaïda [84], we utilized the GJR-GARCH(1,1) model to compute the volatility series. The descriptive statistics for these volatility series are provided in Table 3. As shown in the table, the volatility returns for most assets exhibit positive skewness, indicating that periods of high volatility are more common than those of low volatility. The volatility returns also display exceptionally high kurtosis, especially for crude oil (53.071) and coal (13.407), suggesting that the distributions are leptokurtic with heavy tails. This implies that extreme volatility events are more frequent than would be expected under a normal distribution. The results of the JB test further confirm the non-normality of the volatility returns, with particularly high values, such as 338372.332 for crude oil. This supports the presence of outliers and underscores the non-normal nature of the volatility data. Additionally, the ADF test statistics indicate that the volatility returns are stationary across all assets, suggesting that volatility shocks are temporary and do not exhibit long-term trends. Finally, the results of the Ljung-Box test show strong autocorrelation in the volatility returns for nearly all assets. Crude oil, in particular, exhibits the most significant autocorrelation (Q(5) = 10976.367, Q(20) = 27564.359), indicating that volatility tends to persist over time, with market shocks having lasting effects on price fluctuations.

3.4.1. Static connectedness

This section employs the static network connectedness methodology established by Diebold and Yilmaz [76] to examine the contagion effects between the crude oil futures market and 11 additional markets, including gold, copper, and green energy stock indices. This approach entails estimating a vector autoregressive (VAR) model and subsequently developing connectivity indices derived from the generalized forecast error variance decomposition (GFEVD). These indices allow us to measure the intensity and direction of contagion effects among the variables.

The ideal lag order for the VAR model is established by the Akaike Information Criterion (AIC), indicating a lag order of 2. The VAR(2) model is delineated as $\Lambda_t = \phi\Lambda_{t-1} + \varepsilon_t$, where Λ_t represents the dataset comprising returns or volatilities of crude oil futures and eleven stock

Table 2
Unconditional correlation matrix.

	Crude oil	Natural gas	Solar	Coal	Wind	Gold	Copper	Dow Jones	NASDAQ	US dollar	2y bond	10y bond
Crude oil	1											
Natural gas	0.0942	1										
Solar	0.0713	0.0309	1									
Coal	0.1606	0.0846	-0.0369	1								
Wind	0.0844	0.0155	0.4184	-0.0036	1							
Gold	0.0513	-0.0318	0.0465	0.0147	0.0374	1						
Copper	0.3197	0.0383	0.1421	0.0239	0.1643	0.2548	1					
Dow Jones	0.3421	0.0715	0.0748	-0.0006	0.0908	-0.0401	0.3082	1				
NASDAQ	0.2276	0.0581	0.1075	-0.0127	0.1083	0.0071	0.2989	0.8098	1			
US dollar	-0.0836	0.0006	-0.1105	0.0374	-0.0392	-0.4522	-0.3482	-0.1690	-0.1944	1		
2y bond	-0.0975	-0.0139	-0.0079	-0.0099	-0.0183	0.1844	-0.0070	-0.0569	0.0045	-0.2103	1	
10y bond	-0.1575	-0.0399	0.0064	-0.0091	0.0222	0.3769	-0.0252	-0.1559	-0.0838	-0.2777	0.3927	1

Table 3
Descriptive statistics of volatility return series.

	Mean	Standard deviation	Skewness	Kurtosis	JB	ADF	Q(5)	Q(20)
Crude oil	2.402	1.505	6.066	53.071	338 372.332	-4.605	10 976.367	27 564.359
Coal	3.236	1.890	2.985	13.407	16 033.163	-4.644	8028.018	22 728.438
Solar	1.841	1.319	0.893	0.638	326.793	-2.285	10 423.507	35 853.209
Natural gas	3.570	1.545	1.127	2.663	1128.798	-4.767	9908.444	30 298.410
Wind	1.794	0.580	1.537	2.943	1619.858	-7.510	8043.526	17 510.416
Gold	0.891	0.221	2.155	8.008	7699.193	-4.902	9864.075	28 243.269
Copper	1.366	0.205	1.286	2.632	1260.292	-5.930	9160.700	22 793.696
Dow Jones	0.930	0.662	5.692	48.229	227 365.485	-6.911	9070.164	20 409.105
NASDAQ	1.232	0.642	3.404	21.200	45 906.252	-5.953	9368.611	22 855.893
2 year bond	0.164	0.100	1.516	1.463	1053.617	-8.646	7710.063	15 101.821
10 year bond	0.400	0.097	0.995	1.056	473.013	-4.585	10 285.091	32 246.055
US dollar index	0.405	0.093	1.601	3.473	2078.734	-4.220	10 362.747	33 506.752

indices, and $\varepsilon \sim (0, \Sigma)$ is an independent, identically distributed error term with a mean of 0 and a variance-covariance matrix Σ .

To enable the computation of the GFEVD, the VAR model is transformed into its moving average (MA) representation as $\Lambda_t = \sum_{i=0}^{\infty} \Gamma_i \varepsilon_{t-i}$, where the coefficient matrix Γ_i satisfies: $\Gamma_i = \phi \Gamma_{i-1}$, with Γ_0 as $N \times N$ identity matrix and $\Gamma_i = 0$ for $i < 0$. Then, we perform H-step-ahead GFEVD, setting $H = 100$, following the methodology used in [85,86].

The H-step-ahead GFEVD is denoted by $v_y^g(H)$, for $H = 1, 2, \dots$, calculated as:

$$v_y^g(H) = \frac{\sum_{j=1}^{N-1} \sum_{h=0}^{H-1} (p_i \Gamma_h \Sigma p_j)}{\sum_{h=0}^{H-1} (p'_i \Gamma_h \Sigma \Gamma'_h p_i)} \quad (5)$$

where p_i (p_j) are $N \times 1$ vector with the i th and j th elements set to 1, respectively, while all other elements are set to 0. Using these definitions, we derive the following connectedness indices to analyze the static return and volatility spillovers from variable j to variable i :

$$\tilde{v}_{ij}^g(H) = \frac{v_{ij}^g(H)}{\sum_{i=1, j=1}^N v_{ij}^g(H)} \quad (6)$$

Note that, by construction, $\sum_{j=1}^N \tilde{v}_{ij}^g(H) = 1$ and $\sum_{i=1, j=1}^N \tilde{v}_{ij}^g(H) = N$.

Using Eq. (6), we derive several connectedness indices to perform the static return and volatility spillover analysis. These include the total connectedness index, net directional connectedness index, From connectedness index, To connectedness index, and net pairwise directional connectedness index. The connectedness indices are defined as follows:

The **Total Connectedness Index (TCI)** quantifies the overall spillover intensity among all variables and is calculated as:

$$\Lambda^g(H) = \frac{\sum_{i=1, j=1}^N \tilde{v}_{ij}^g(H)}{\sum_{i=1, j=1}^N \tilde{v}_{ij}^g(H)} \times 100 = \frac{\sum_{i=1, j=1}^N \tilde{v}_{ij}^g(H)}{N} \times 100. \quad (7)$$

This index measures the proportion of variance in the forecast errors of all variables that can be attributed to spillovers from other variables, thereby providing a comprehensive view of the interconnectedness across the system.

The **From Connectedness Index** measures the spillover effects received by variable i from all other variables j , which can be calculated

as:

$$\Lambda_{i \cdot}^g(H) = \frac{\sum_{j=1}^N \tilde{v}_{ji}^g(H)}{\sum_{i=1, j=1}^N \tilde{v}_{ij}^g(H)} \times 100 = \frac{\sum_{j=1}^N \tilde{v}_{ji}^g(H)}{N} \times 100. \quad (8)$$

This index reflects the extent to which the variability in variable i 's forecast errors is explained by shocks originating from all other variables, thereby highlighting its sensitivity to external influences.

The **To Connectedness Index** quantifies the spillover effects transmitted from variable i to all other variables j , which can be calculated as:

$$\Lambda_{\cdot i}^g(H) = \frac{\sum_{j=1, j \neq i}^N \tilde{v}_{ji}^g(H)}{\sum_{i=1, j=1}^N \tilde{v}_{ij}^g(H)} \times 100 = \frac{\sum_{j=1, j \neq i}^N \tilde{v}_{ji}^g(H)}{N} \times 100 \quad (9)$$

This index measures the contribution of variable i to the variance in the forecast errors of other variables, highlighting its role as a transmitter of spillovers.

The **Net Directional Connectedness Index** measures the net spillover effect for variable i , capturing the difference between the spillover effects transmitted by i to all other variables and the spillover effects received by i from all other variables. It is calculated as:

$$\Lambda_i^g(H) = \Lambda_{\cdot i}^g(H) - \Lambda_{i \cdot}^g(H) \quad (10)$$

This index indicates whether a variable acts primarily as a transmitter ($C_i^{\text{Net}} > 0$) or a receiver ($C_i^{\text{Net}} < 0$) of spillover effects within the system.

The **Net Pairwise Directional Connectedness Index** captures the directional spillover between two variables i and j . It is calculated as:

$$\begin{aligned} \Lambda_{ij}^g(H) &= \left(\frac{\tilde{v}_{ji}^g(H)}{\sum_{i=1, j=1}^N \tilde{v}_{ij}^g(H)} - \frac{\tilde{v}_{ij}^g(H)}{\sum_{j=1, i=1}^N \tilde{v}_{ij}^g(H)} \right) \times 100 \\ &= \left(\frac{\tilde{v}_{ji}^g(H)}{N} - \frac{\tilde{v}_{ij}^g(H)}{N} \right) \times 100. \end{aligned} \quad (11)$$

This index provides insights into the dominant direction of spillover effects between two specific variables, indicating whether variable i transmits more information to variable j , or vice versa.

Table 4
Static volatility connectedness.

	Crude oil	Natural gas	Solar	Coal	Wind	Gold	Copper	Dow Jones	NASDAQ	US dollar	2y bond	10y bond	From
Crude oil	47.8209	0.0763	1.2232	1.4705	3.1236	8.3527	2.4130	8.2181	18.3938	0.0169	0.3692	8.5219	4.3483
Natural gas	2.2592	75.7090	2.1622	2.5190	6.0177	0.8590	0.1080	6.3344	1.8958	0.0327	0.7138	1.3893	2.0242
Solar	0.6543	6.7504	80.6138	2.9368	0.8341	0.6392	0.2578	4.2337	0.6570	0.0228	0.0605	2.3395	1.6155
Coal	2.6160	4.7588	0.9512	83.3589	2.8645	2.2004	1.5577	0.2906	0.3791	0.1226	0.2884	0.6117	1.3868
Wind	1.4763	2.9767	10.5969	3.3994	72.5142	3.6939	0.3835	2.9485	0.3500	0.0139	0.0278	1.6189	2.2905
Gold	0.2137	1.0430	5.4490	2.3623	1.9572	60.2272	0.2099	11.3489	9.4042	0.0317	1.4781	6.2749	3.3144
Copper	2.2125	5.8181	2.2072	2.5912	3.9096	0.2798	62.7359	8.0625	11.6510	0.0539	0.2371	0.2412	3.1053
Dow Jones	0.5399	0.2150	0.7281	0.2699	0.2571	2.6944	0.4222	92.4935	1.6144	0.0125	0.0519	0.7010	0.6255
NASDAQ	0.2293	0.8272	0.2932	2.2784	7.2904	0.1690	1.9705	18.6646	64.7950	0.0450	1.0336	2.4039	2.9337
US dollar	0.6612	1.0328	2.3087	0.2579	13.7292	0.7654	1.2821	5.3663	1.1907	70.3931	0.0693	2.9434	2.4672
2y bond	0.2706	2.6843	0.9308	0.4369	0.4661	0.9741	0.1546	0.6237	0.8247	0.0030	89.8549	2.7763	0.8454
10y bond	1.0203	2.7243	0.2166	19.8771	2.7617	0.3691	0.3087	4.8446	5.3200	0.1012	2.0152	60.4414	3.2966
To	1.0128	2.4089	2.2556	3.2000	3.6009	1.7498	0.7557	5.9113	4.3067	0.0380	0.5287	2.4852	Total
Net	-40.0259	4.6159	7.6808	21.7585	15.7255	-18.7757	-28.1961	63.4291	16.4757	-29.1508	-3.8002	-9.7368	28.2535

To better understand the interconnectedness between the crude oil market and the other 11 markets, we analyze the static volatility connectedness indices, based on the network connectedness method proposed by Diebold and Yilmaz [76]. The results of the static volatility connectedness indices are presented in Table 4. Firstly, the total volatility connectedness index among the 12 markets is 28.2535%, indicating a moderate level of volatility spillover across these markets. This value reflects the proportion of total volatility that can be attributed to spillover effects rather than market-specific factors. The “To” connectedness indices highlight the extent to which each market contributes to volatility spillovers in other markets. Notably, the NASDAQ (5.9113%) and the crude oil market (1.0128%) are among the largest contributors to volatility transmission. The “From” connectedness indices indicate the extent to which each market is influenced by volatility spillovers from others. Crude oil receives the highest “From” index at 47.8209%, underscoring its sensitivity to external shocks from other markets, particularly from financial markets like the Dow Jones and NASDAQ. Importantly, the Net directional connectedness indices provide further insight. Crude oil is a net receiver of volatility spillover effects, with a net index of -40.0259, demonstrating that it is more affected by other markets than it influences them. In summary, the crude oil market is heavily influenced by external volatility spillovers, particularly from financial markets, while its impact on other markets remains limited.

Based on Table 4, we have draw Fig. 4 From and To pairwise volatility network based on static connectedness. The arrows represent the direction of spillover effects between crude oil (WTI) and other assets, while the color intensity reflects the strength of the connectedness, ranging from weak (blue) to strong (orange). For instance, crude oil exhibits strong spillover effects with key markets such as natural gas, Dow Jones Industrial, and NASDAQ.

Regarding weak connectedness (illustrated by blue arrows in Fig. 4), it is important to emphasize that these linkages, although modest in magnitude, represent statistically significant and non-zero volatility relationships. Such weak connections are a characteristic feature of complex financial and commodity markets, as evidenced by recent academic studies. For instance, Zhang et al. [87] demonstrated that the US dollar exchange rate exerts a long-term equilibrium effect on crude oil prices, albeit with limited short-term volatility spillovers, underscoring the presence of weak but persistent linkages. Similarly, Shang and Hamori [88] revealed that crude oil consistently transmits shocks across various quantiles and time frequencies to gold and financial markets, indicating the systemic importance of even weak connections. Mensi et al. [89] further highlighted the time-varying nature of weak connectedness between oil and stock markets, showing that these linkages intensify during crises, thereby providing valuable insights for portfolio management. Asadi et al. [90] extended this analysis to include natural gas, coal, and currency markets, confirming the existence of statistically significant weak spillovers across multiple asset classes. Guo and Zhao [91] specifically examined volatility spillovers between oil and coal prices in China, finding significant weak connections that have implications for energy portfolio diversification. Perifanis and

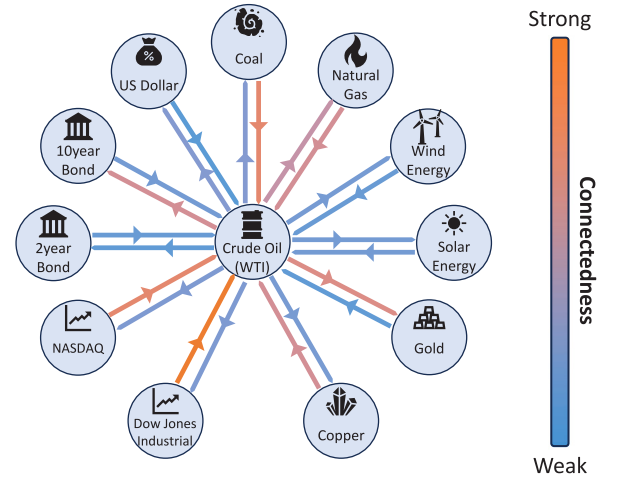


Fig. 4. From and to pairwise volatility network based on static connectedness.

Dagoumas [92] identified bidirectional volatility transmission between US crude oil and natural gas markets, albeit at a weak level, suggesting that market fundamentals drive price formulation despite decoupled returns. Moreover, Alharbey et al. [93] reported asymmetric volatility spillovers between oil prices and regional renewable energy stock markets, with weak connections varying by region and becoming more pronounced during major events. Collectively, these studies validate the presence and significance of weak connectedness, as depicted by blue arrows in Fig. 4, emphasizing their role in understanding market dynamics and informing risk management strategies.

3.4.2. Dynamic connectedness

This section utilizes the dynamic network connectedness framework developed by Antonakakis et al. [94] to investigate the time-varying spillover effects between the crude oil futures market and 11 additional variables. These variables include gold, copper, green energy indices, stock markets, bond markets, and other critical economic indicators. The approach involves estimating the coefficients of a time-varying parameter vector autoregressive (TVP-VAR) model using the Kalman filter technique. The results from the GFEVD are then used to construct dynamic connectedness indices, which quantify the strength and direction of spillover effects among variables over time.

The estimation of the dynamic coefficients begins with the TVP-VAR model, which is defined as $A_t = B_t A_{t-1} + u_t$, where $u_t \sim N(0, \Sigma_t)$, where, $\text{vec}(B_t) = \text{vec}(B_{t-1}) + v_t$, $v_t \sim N(0, \Phi_t)$, A_t is a $N \times 1$ vector representing observations of N variables (e.g., returns or volatilities of crude oil futures and the other 11 factors) at time t , B_t is a $N \times N$ time-varying coefficient matrix, $\text{vec}(B_t)$ is the vectorized form of B_t , u_t is the error term following a normal distribution with variance-covariance matrix Σ_t , v_t is the error term for the time-varying coefficients, following

a normal distribution with variance–covariance matrix Φ_t . Then, The Kalman filter is employed to estimate the dynamic coefficients B_t and the variance–covariance matrix Σ_t of the model. For detailed steps on the implementation of the Kalman filter, refer to Antonakakis et al. [94].

Similarly, the TVP-VAR model is reformulated as a time-varying parameter vector moving average (TVP-VMA) model to facilitate the analysis, $A_t = B_t A_{t-1} + u_t = \sum_{j=0}^{\infty} \Gamma_{jt} u_{t-j}$, where, Γ_{jt} represents the time-varying moving average coefficients, which facilitate the application of the GFEVD. GFEVD is then employed to quantify the contribution of shocks from one variable to the forecast error variance of another. The H-step-ahead GFEVD at time t , according to Eq. (5), is expressed as:

$$v_{jk,t}^g(H) = \frac{\sum_{k=1}^N \sum_{h=0}^{H-1} \left(p'_{j,t} \Gamma_{h,t} \Sigma_t p_{k,t} \right)^2}{\sum_{h=0}^{H-1} \left(p'_{j,t} \Gamma_{h,t} \Sigma_t \Gamma'_{h,t} p_{j,t} \right)}, \quad (12)$$

where, $p_{j,t}$ and $p_{k,t}$ are selection vectors with the j th and k th elements set to 1, respectively, while all other elements are set to 0. Σ_t is the time-varying variance–covariance matrix. After standardizing the GFEVD, the dynamic connectedness indices are constructed as follows:

$$\tilde{v}_{jk,t}^g(H) = \frac{v_{jk,t}^g(H)}{\sum_{j=1,k=1}^N v_{jk,t}^g(H)}. \quad (13)$$

These indices satisfy the conditions $\sum_{k=1}^N \tilde{v}_{jk,t}^g(H) = 1$ and $\sum_{j=1,k=1}^N \tilde{v}_{jk,t}^g(H) = N$.

Utilizing Eq. (13), the dynamic total connectedness index, dynamic net directional connectedness index, dynamic From connectedness index, dynamic To connectedness index as well as the dynamic net pairwise directional connectedness index are constructed as follow:

Total connectedness index:

$$\lambda_t^g(H) = \frac{\sum_{j=1,k=1}^N \tilde{v}_{jk,t}^g(H)}{\sum_{j=1,k=1}^N \tilde{v}_{jk,t}^g(H)} \times 100 = \frac{\sum_{j=1,k=1}^N \tilde{v}_{jk,t}^g(H)}{N} \times 100 \quad (14)$$

From connectedness index:

$$\lambda_{j,t}^g(H) = \frac{\sum_{k=1}^N \tilde{v}_{jk,t}^g(H)}{\sum_{j=1,k=1}^N \tilde{v}_{jk,t}^g(H)} \times 100 = \frac{\sum_{k=1}^N \tilde{v}_{jk,t}^g(H)}{N} \times 100. \quad (15)$$

To connectedness index:

$$\lambda_{\cdot,j,t}^g(H) = \frac{\sum_{k=1}^N \tilde{v}_{kj,t}^g(H)}{\sum_{j=1,k=1}^N \tilde{v}_{kj,t}^g(H)} \times 100 = \frac{\sum_{k=1}^N \tilde{v}_{kj,t}^g(H)}{N} \times 100. \quad (16)$$

Net directional connectedness index:

$$\lambda_{j,t}^g(H) = S_{j,t}^g(H) - S_{\cdot,j,t}^g(H). \quad (17)$$

Net pairwise directional connectedness index:

$$\begin{aligned} \lambda_{jk,t}^g(H) &= \left(\frac{\tilde{v}_{kj,t}^g(H)}{\sum_{j=1,k=1}^N \tilde{v}_{jk,t}^g(H)} - \frac{\tilde{v}_{jk,t}^g(H)}{\sum_{j=1,k=1}^N \tilde{v}_{jk,t}^g(H)} \right) \times 100 \\ &= \left(\frac{\tilde{v}_{kj,t}^g(H)}{N} - \frac{\tilde{v}_{jk,t}^g(H)}{N} \right) \times 100. \end{aligned} \quad (18)$$

To analyze the time-varying connectedness index and their evolution over time, Fig. 5 illustrates the Dynamic Total Return Connectedness Index over the sample period. This figure provides insights into the time-varying spillover intensity among the analyzed variables, highlighting how connectedness evolves dynamically over time. A sharp peak is observed around 2020, corresponding to significant market turbulence likely triggered by the COVID-19 pandemic and associated economic uncertainties. This peak indicates heightened spillover intensity across markets during periods of financial stress. The dynamic total return connectedness index captures the evolving nature of spillover effects among the crude oil futures market and the other variables.

The selection of the optimal lag order in the VAR model, as guided by the methodology in [95], involves the use of information criteria to balance model complexity and goodness of fit. The Akaike Information Criterion (AIC) is instrumental in this process, with lower AIC values indicating models that effectively explain the data while minimizing overfitting. Among the models evaluated, Fig. 6(b) emerges as the optimal choice. This model achieves a balance between reduced complexity and effective capture of dynamic changes, characterized by stable fluctuations. Such stability suggests the model's proficiency in representing long-term trends and periodic changes without succumbing to overfitting, thereby achieving a lower AIC value. In contrast, Fig. 6(a) exhibits high-frequency fluctuations, which, while adept at capturing short-term dynamics, increase model complexity and AIC values. Fig. 6(c) is capable of capturing long-term trends but does so at the expense of increased complexity. Fig. 6(d) shows significant sensitivity to outliers or abrupt events, which enhances complexity and potentially elevates AIC values. Fig. 6(e) demonstrates minor fluctuations and an overall stable trend, indicating robustness but possibly a conservative approach in capturing dynamic changes. Meanwhile, Fig. 6(f) captures periodic changes effectively, yet these fluctuations might lead to increased complexity and higher AIC values.

Overall, Fig. 6(b) is distinguished by its ability to capture dynamic changes with reduced complexity and superior fit quality, making it the most balanced and effective model among those considered. Thus, we choose lag order = 2 for the TVP-VAR model in this work.

3.5. The proposed RLDC-LLM framework

The architecture of our proposed RLDC-LLM framework, which uses Proximal Policy Optimization (PPO) to improve WTI crude oil futures trading performance, is shown in Fig. 7. The twelve assets that were chosen for the framework are combined with insights from the static and dynamic connectivity analyses. The analyzed issue is articulated as a Markov Decision Process (MDP), expressed as a 5-tuple. $(S, \mathcal{A}, \mathcal{P}(\cdot), \mathcal{R}(\cdot), \gamma)$, where S represents the set of states, $\mathcal{A}$ signifies the action space, and $\mathcal{P}(\cdot) : S \times \mathcal{A} \times S \rightarrow [0, 1]$ represents the state transition probability density function characterizing the market environment, $\mathcal{R}(\cdot) : S \times \mathcal{A} \rightarrow \mathbb{R}$ defines the reward signal, and $\gamma \in [0, 1]$ signifies the discount factor. The agent aims to investigate a strategy π that maximizes the cumulative discounted reward, expressed as $\sum_{t=0}^T \gamma^t \mathcal{R}(s_t, a_t)$. Specifically, we possess:

3.5.1. State

The state features used in the RL-based trading strategy include various combinations of fundamental market information, sentiment metrics, disagreement metrics, cross-market features, and connectedness metrics. Each state $s_t \in S$ is designed to represent the information available to the RL agent at time t . Below, the specific state configurations, their purposes, and their formulas are explained:

- **Basic Market State:** The basic state consists of the fundamental market features for various assets X , providing essential price and volume information combined with the agent's current position (hold state). It is represented as:

$$s_t^X = \{\mathcal{O}_t^X, \mathcal{H}_t^X, \mathcal{L}_t^X, \mathcal{C}_t^X, \mathcal{V}_t^X\} \cup \kappa_t, \quad (19)$$

where, $\mathcal{O}_t^X, \mathcal{H}_t^X, \mathcal{L}_t^X, \mathcal{C}_t^X, \mathcal{V}_t^X$, are the open, high, low, close prices, and volume, respectively. κ_t represents the hold state for WTI crude oil futures, which indicates the current position size of the RL agent. This component ensures that the agent considers its existing exposure when making trading decisions. For example, the basic market state for WTI crude oil, denoted as s_t^{WTI} , where s_t^{WTI} can be represented as: $s_t^{\text{WTI}} = \{\mathcal{O}_t^{\text{WTI}}, \mathcal{H}_t^{\text{WTI}}, \mathcal{L}_t^{\text{WTI}}, \mathcal{C}_t^{\text{WTI}}, \mathcal{V}_t^{\text{WTI}}\} \cup \kappa_t$. This configuration serves as the foundation for more advanced state representations.

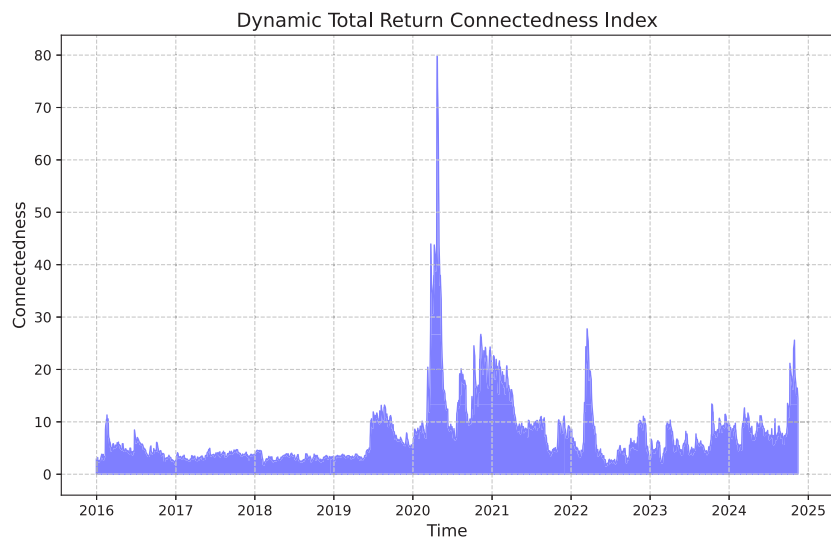
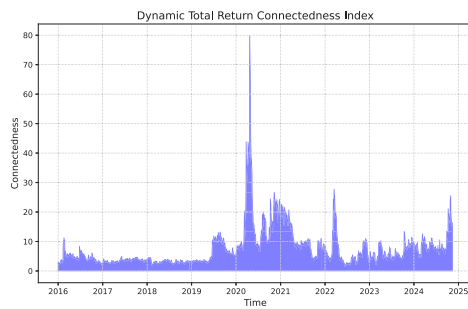
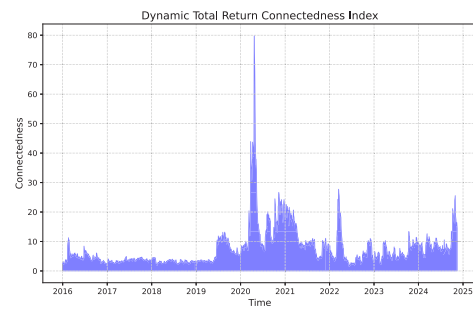


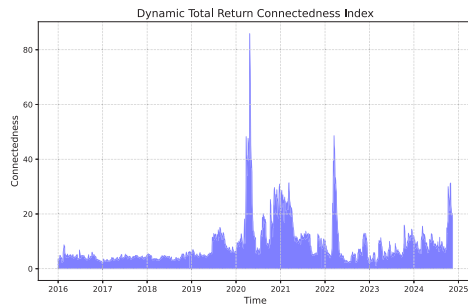
Fig. 5. Net pairwise volatility spillover network based on dynamic connectedness.



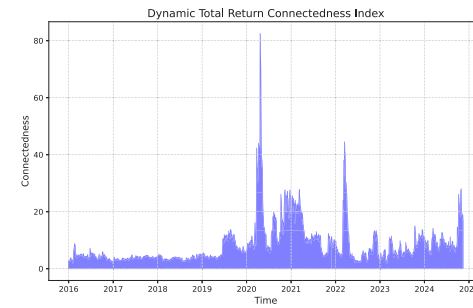
(a) lag order = 1



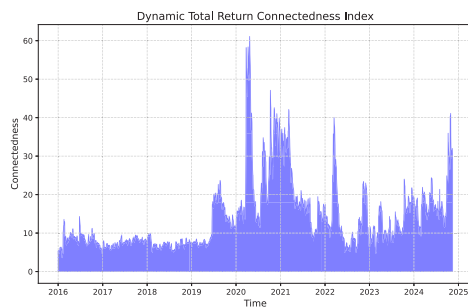
(b) lag order = 2



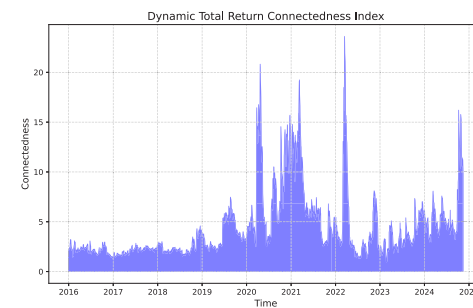
(c) lag order = 3



(d) lag order = 4



(e) lag order = 5



(f) lag order = 6

Fig. 6. Dynamic connectedness using different lag order settings.

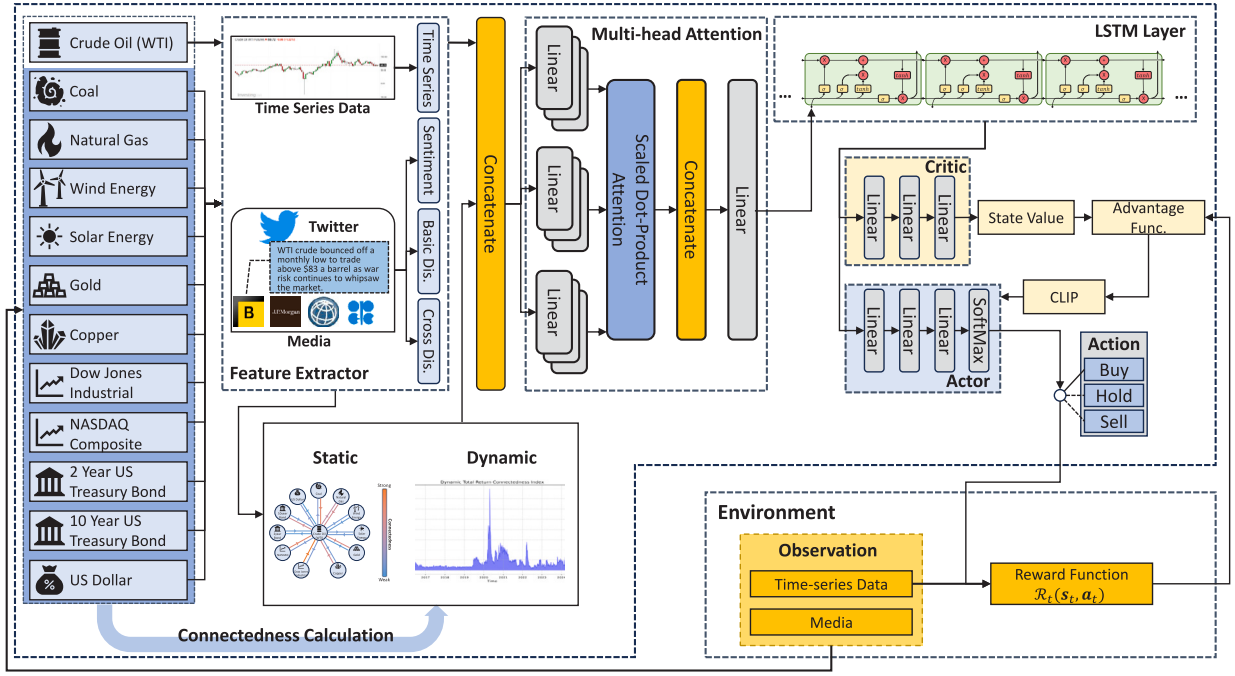


Fig. 7. It consists of twelve assets contained with the WTI to perceive the four different features from the proposed feature extractor, i.e. time series feature, sentiment feature, basic disagreement feature, and cross disagreement feature. Except for the WTI, the other eleven assets' features are calculated by static or dynamic connectedness to enhance the correlation with WTI trading, all twelve assets are then concatenated together as the input for the PPO module. The multi-head attention module and the LSTM Layer are used to capture the critical features from the observation features. The actor and critic networks share the same architecture. The main difference is that the output nodes for the actor correspond to the three actions "buy", "hold", and "sell", and the output node for the critic is to estimate the state value. A well-designed reward with profit or loss and the transaction cost incurred is built to guide the training of the agent.

- **Cross-Market State:** The Cross-Market State includes fundamental market features from external markets (e.g., gold, natural gas) to capture inter-market dependencies:

$$s_t^{X+Y} = s_t^X \cup \{\mathcal{O}_t^Y, H_t^Y, L_t^Y, C_t^Y, V_t^Y\}, \quad (20)$$

where, Y represents an external market (e.g., gold, natural gas), $\mathcal{O}_t^Y$, H_t^Y , L_t^Y , C_t^Y , and V_t^Y denote the open, high, low, close prices, and trading volume of the external asset Y at time t , respectively. For example:

$$-Y = \text{Natural Gas: } s_t^{\text{WTI}+\text{NatGas}} = s_t^{\text{WTI}} \cup \{\mathcal{O}_t^{\text{NatGas}}, H_t^{\text{NatGas}}, L_t^{\text{NatGas}}, C_t^{\text{NatGas}}, V_t^{\text{NatGas}}\}.$$

This state allows the RL agent to consider interdependencies between WTI crude oil and external assets for better trading decisions.

- **Disagreement-Driven State:**

The Disagreement-Driven State first augments the basic market state by incorporating market sentiment. Sentiment is derived from media reports and social media interactions, including a temporal decay mechanism to prioritize recent sentiments. Then, it extends the sentiment-enhanced by adding the degree of disagreement among market participants, which reflects market uncertainty. The disagreement-driven state for a single asset X is defined as:

$$s_t^{X+D^b} = s_t^X \cup \{D_{X,t}^b\}, \quad (21)$$

where $D_{X,t}^b$ denotes the basic disagreement metric for asset X , capturing the variance in opinions among investors about the future price movements of X . This disagreement metric provides critical information about market uncertainty, enabling the RL agent to adapt its strategies based on differing investor outlooks. Moreover, the disagreement-driven state can be further extended to incorporate cross-market features by including disagreement metrics for multiple assets. For instance, when considering an

external market Y in addition to asset X , the state can be represented as:

$$s_t^{X+Y+D^b} = s_t^{X+Y} \cup \{D_{X,t}^b\} \cup \{D_{Y,t}^b\}, \quad (22)$$

where, Y represents an external market (e.g., gold, copper). In this case, $D_{Y,t}^b$ is the basic disagreement metric for market Y , capturing variance in investor opinions regarding Y .

By including disagreement metrics for both X and Y , the RL agent can better understand the uncertainties and differing investor outlooks across multiple markets. This enhanced state representation allows the agent to make more informed trading decisions by considering both market-specific and cross-market uncertainties.

- **Cross-Disagreement State:**

The Cross-Disagreement State incorporates cross-category disagreement metrics among different investor groups (e.g., technical vs. fundamental analysts), providing deeper insights into market uncertainty. By capturing variances in opinions across distinct groups, this state allows the RL agent to gain a more granular understanding of market dynamics and investor behavior. The cross disagreement state for a single asset X is defined as:

$$s_t^{X+D^c} = s_t^X \cup \{D_{X,at}^c\}, \quad (23)$$

where, $D_{X,at}^c$ denotes the cross-disagreement metric for asset X , quantifying differences in opinions between specific investor groups (e.g., group a) regarding asset X .

The cross-disagreement state can be further extended to include cross-market disagreement metrics, thereby capturing the interaction of disagreement metrics across multiple assets. For instance, when considering an external market Y in addition to asset X , the state can be represented as:

$$s_t^{X+Y+D^c} = s_t^{X+Y} \cup \{D_{X,at}^c\} \cup \{D_{Y,at}^c\} \cup \{D_{X,Yt}^c\}, \quad (24)$$

where, Y represents an external market (e.g., gold, copper). In this case, $D_{Y,t}^c$ is the cross investor group disagreement metric

for market Y within group α , capturing variance in investor opinions regarding Y . $D_{X,\beta t}^c$ is the cross market disagreement metric for asset X between different markets (X and Y), reflecting disagreements across markets.

This extended state representation ensures the RL agent can analyze and respond to complex interdependencies and opinion divergences, improving its ability to navigate uncertain and volatile market conditions effectively.

- **Dual Disagreement State:** The Dual Disagreement State builds upon the disagreement-driven state by integrating both basic disagreement (D^b) and cross-disagreements (D^c) among investors and markets. This enriched state representation provides a more comprehensive view of market uncertainty and investor behavior. The dual-disagreement state for a single asset X is defined as:

$$s_t^{X+D^b+D^c} = s_t^{X+D^b} \cup \{D_{X,t}^c\}, \quad (25)$$

where, $D_{X,t}^b$ represents basic disagreement metric for asset X , capturing variance in opinions within a single investor group. $D_{X,t}^c$ represents cross-disagreement metric for investor group X at time t , quantifying differences in opinions across distinct investor categories (e.g., technical vs. fundamental analysts).

The dual-disagreement state can be further extended to incorporate disagreement metrics across multiple markets, thereby capturing the interaction of disagreement effects among different assets. For example, when considering an external market Y in addition to asset X , the state can be represented as:

$$s_t^{X+Y+D^b+D^c} = s_t^{X+Y+D^b} \cup \{D_{X,\alpha t}^c\} \cup \{D_{Y,\alpha t}^c\} \cup \{D_{X,\beta t}^c\}, \quad (26)$$

where, Y represents external market (e.g., gold, copper). $D_{X,\alpha t}^c$ represents cross-disagreement metric for asset X within investor group α , capturing differences of opinions specific to X . $D_{Y,\alpha t}^c$ represents cross-disagreement metric for market Y within investor group α , reflecting differences of opinions for Y . $D_{X,\beta t}^c$ represents cross-market disagreement metric for asset X across markets X and Y , representing differences in opinions between the markets. This state configuration serves as a vital component for developing advanced trading strategies that effectively account for investor behavior and asset interdependencies, making it particularly useful in multi-asset and interconnected markets.

- **Connectedness-based State:** The Connectedness-Based State combines all previously defined features with connectedness metrics to capture interdependencies among crude oil and external markets:

$$s_t^{All+D^b+D^c} = s_t^{X+\sum_{y=1}^{11} Y+D^b+D^c} \cup \{\lambda_t^g(H)\}, \quad (27)$$

where $\lambda_t^g(H)$ represents the connectedness metric for crude oil and external markets.

This comprehensive state enables the RL agent to analyze the relationships and influences among various market segments, such as the interplay between crude oil and green energy markets.

3.5.2. Action

The action space $\mathcal{A}$ in the RL framework is defined as the set of possible operations for trading WTI crude oil futures. Formally, the action at time t is represented as: $\mathbf{a}_t \in \{-1, 0, +1\}$, where:

- $\mathbf{a}_t = +1$: Indicates a decision to buy (long position). Specifically, entering or increasing a long position in the WTI crude oil futures contract, anticipating a price increase.
- $\mathbf{a}_t = -1$: Indicates a decision to sell (short position). Specifically, entering or increasing a short position in the WTI crude oil futures contract, anticipating a price decrease.
- $\mathbf{a}_t = 0$: Indicates a decision to hold the current position. Specifically, maintaining the current position without initiating any new trades, typically in uncertain market conditions.

The RL agent selects actions $\mathbf{a}_t$ at each time step based on the current state $\mathbf{s}_t$, aiming to maximize cumulative rewards over time. This action selection is governed by the learned policy π_θ , which maps states to action probabilities. Specifically, the probability of choosing action $\mathbf{a}_t$ in state $\mathbf{s}_t$ is given by:

$$\pi_\theta(\mathbf{a}_t | \mathbf{s}_t) = \mathbb{P}(\mathbf{a}_t | \mathbf{s}_t), \quad (28)$$

where π_θ is parameterized by θ . The policy ensures that the agent dynamically adapts its actions to evolving market conditions, balancing risks and returns.

3.5.3. Reward

The reward function in this study is defined as the net profit achieved by the RL agent during each trading step, adjusted for transaction costs. Specifically, the reward is calculated as the realized profit or loss from the trading action taken at time t , minus a transaction fee set at 0.5% of the traded amount. This design ensures that the RL agent not only aims to maximize returns but also considers the impact of trading costs, encouraging more efficient and realistic trading behavior. Formally, the reward function is expressed as:

$$\mathcal{R}_t = \mathcal{K}_t - F_t, \quad (29)$$

where, $\mathcal{K}_t$ denotes the profit or loss generated from the trading action at time t , calculated as the change in asset price multiplied by the position size, and F_t denotes the transaction cost incurred. The transaction cost incurred defined as $F_t = 0.005 \times \mathcal{E}_t$, where $\mathcal{E}_t$ is the monetary value of the asset bought or sold at time t .

Reward shaping is not utilized in the design of our RLDC-LLM framework primarily to maintain simplicity and clarity in the reward function. By avoiding complex modifications to the reward structure, our framework reduces the risk of introducing biases that could lead to suboptimal policy learning. This approach ensures that the reward function remains closely aligned with realistic trading conditions, where rewards are directly linked to actual trading profits and transaction costs. This direct linkage helps prevent the distortion of real-world signals, which could occur with reward shaping, resulting in strategies that might perform well in training data but fail in testing data. Furthermore, avoiding reward shaping minimizes the potential for overfitting to specific patterns in the training data, thereby promoting the development of robust and generalizable trading strategies.

3.5.4. State transition

State transitions occur as the market evolves over time. At each time step t , the state $\mathbf{s}_t$ transitions to the next state $\mathbf{s}_{t+1}$ based on the observed market movements and the agent's actions $\mathbf{a}_t$. The transition dynamics are influenced by factors such as asset price changes, sentiment updates, and connectedness spillovers. The state transition from $\mathbf{s}_t$ to $\mathbf{s}_{t+1}$ is governed by the function $\mathbf{s}_{t+1} = F(\mathbf{s}_t, \mathbf{a}_t, F_t)$. The transition may not only depend on the action $\mathbf{a}_t$ but can also be affected by the uncertainties F_t that exist in some state features.

3.5.5. Policy-based reinforcement learning

Proximal Policy Optimization (PPO) [22] is a prevalent on-policy deep reinforcement learning method extensively applied across diverse fields such as robotics [96], planning [97], pricing [98,99] and portfolio optimization [100]. It aims to train policies that optimize rewards while maintaining stability and efficiency in learning. PPO computes updates at each iteration to maximize the cost function while reducing the divergence between the new and previous policies. This characteristic markedly diminishes the necessity for extensive parameter tuning, leading to a more streamlined and effective optimization procedure. This work utilizes the PPO algorithm, which is based on a Gaussian distribution, to address continuous action spaces, aligning with the continuous-time nature of crude oil trading features. This approach

allows the model to effectively capture the dynamic and fluid characteristics of trading, enabling precise and adaptive decision-making in real-time market conditions.

A policy $\pi_\theta : S \times \mathcal{A} \rightarrow [0, 1]$ is a parameterized mapping from state space to action space. To address the problem of excessively large policy updates in the PPO algorithm, we incorporate the clipped surrogate objective, which is expressed as:

$$\mathcal{L}(\theta) = \mathbb{E}_{\pi_\theta} [\min(k_t(\theta)\mathcal{A}^{\pi_\theta}, \text{clip}(k_t(\theta), 1 - \epsilon, 1 + \epsilon)\mathcal{A}^{\pi_\theta})] \quad (30)$$

where, $k_t(\theta)$ denotes the probability ratio, which can be computed by:

$$k_t(\theta) = \frac{\pi_\theta(s_t, \mathbf{a}_t)}{\pi_{\theta_{\text{old}}}(s_t, \mathbf{a}_t)}, \quad (31)$$

and $\mathcal{A}^{\pi_\theta}$ denoted the consequence advantage function with mapping as $S \rightarrow \mathbb{R}$, which is expressed as:

$$\mathcal{A}^{\pi_\theta}(s_t, \mathbf{a}_t) = \mathcal{Q}^{\pi_\theta}(s_t, \mathbf{a}_t) - \mathcal{V}^{\pi_\theta}(s_t) - \alpha \log p^{\pi_\theta}(\mathbf{a}_t | s_t), \quad (32)$$

in which, $\mathcal{Q}^{\pi_\theta} : S \times \mathcal{A} \rightarrow \mathbb{R}$ denotes the corresponding action-value function (Q-function), which is defined as:

$$\mathcal{Q}^{\pi_\theta}(s_t, \mathbf{a}_t) = \mathbb{E} \left[\sum_{n=0}^{\infty} \gamma^n \cdot \mathcal{R}(s_t, \mathbf{a}_t) \mid s_t = s, \mathbf{a}_t = \mathbf{a} \right], \quad (33)$$

$\mathcal{V}^{\pi_\theta} : S \rightarrow \mathbb{R}$ denotes the state-value function, which is defined as:

$$\mathcal{V}^{\pi_\theta}(s_t) = \mathbb{E} \left[\sum_{n=0}^{\infty} \gamma^n \cdot \mathcal{R}(s_t, \mathbf{a}_t) \mid s_t = s \right], \quad (34)$$

α denotes the coefficient that balances the influence of the log probability term in the advantage function, and $p^{\pi_\theta}(\mathbf{a}_t | s_t)$ denotes the Gaussian distribution parameterized with mean $\mu^{\pi_\theta}(s_t)$, standard deviation $\sigma^{\pi_\theta}(s_t)$ and circular constant Π , which can be computed by:

$$p^{\pi_\theta}(\mathbf{a}_t | s_t) = \frac{1}{\sqrt{2\Pi\sigma^{\pi_\theta}(s_t)^2}} \exp \left(-\frac{\mathbf{a}_t - \mu^{\pi_\theta}(s_t)^2}{2\sigma^{\pi_\theta}(s_t)^2} \right). \quad (35)$$

3.5.6. Multi-head attention

The multi-head attention mechanism, introduced by Vaswani et al. [101], is a powerful technique for representation learning that has been widely adopted in various domains due to its ability to capture complex dependencies within data. Unlike single-head attention, which focuses on a single aspect of the input, multi-head attention employs multiple parallel attention heads, each learning distinct patterns and relationships in the data. This mechanism allows the model to attend to diverse and complementary features, such as temporal dependencies, cross-market interactions, and correlations among state features.

In this study, the state features, which include connectedness metrics, sentiment signals, disagreement measures, and cross-market interactions, exhibit high dimensionality and complex relationships that demand robust feature extraction. Multi-head attention dynamically assigns relevance weights to these features at each time step, enabling the RL agent to focus on the most critical signals, such as shifts in connectedness or sentiment, while attending to multiple time steps and interdependencies simultaneously.

Mathematically, the output of multi-head attention is defined as:

$$\text{MultiHead}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{Concat}(\text{head}_1, \text{head}_2, \dots, \text{head}_h) \mathbf{W}^O, \quad (36)$$

where each attention head is computed as:

$$\text{head}_i = \text{Attention}(\mathbf{Q}\mathbf{W}_i^Q, \mathbf{K}\mathbf{W}_i^K, \mathbf{V}\mathbf{W}_i^V), \quad (37)$$

and the scaled dot-product attention function is defined as:

$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax} \left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d_k}} \right) \mathbf{V}. \quad (38)$$

Here, $\mathbf{Q}$ (queries), $\mathbf{K}$ (keys), and $\mathbf{V}$ (values) are derived from the input features, $\mathbf{W}_i^Q$, $\mathbf{W}_i^K$, and $\mathbf{W}_i^V$ are learnable projection matrices for the i th attention head, d_k is the dimension of the key vector, and the scaling factor $\sqrt{d_k}$ prevents large dot-product values that

could destabilize the softmax operation, $\mathbf{W}^O$ is a learnable matrix that combines the outputs of all attention heads.

By incorporating multi-head attention into the state feature extraction process, this study enhances the RL agent's ability to trade crude oil. The output of the mechanism is expressed as:

$$\mathbf{h}_t = \text{MultiHead}(\mathbf{Q}_t, \mathbf{K}_t, \mathbf{V}_t), \quad (39)$$

which serves as an enriched state representation for the RL policy network. This enables more effective decision-making in volatile and dynamic market environments.

The incorporation of a multi-head attention mechanism enables the model to dynamically focus on the most relevant features at each time step, thus maintaining responsiveness to changes in the importance of different signals over time. Additionally, the integration of disagreement metrics helps the model capture evolving investor sentiment and opinion divergence, providing insights into changing market dynamics. By leveraging LLMs, our framework enhances its ability to process unstructured textual data, such as news and social media, which are inherently non-stationary. This capability allows the model to incorporate new information and adjust to emerging trends, ensuring its adaptability in the real world dynamic markets.

4. Experimental results

4.1. Data

The market data we obtain is from the WIND database. Apart from West Texas Intermediate (WTI), we also select 10 different assets that may have an influence on WTI, including commodities such as Coal, Copper, Gold, and Natural Gas; Treasury Note Futures, specifically CBOT2 (Chicago Board of Trade 2-Year Treasury Note Futures) and CBOT10 (Chicago Board of Trade 10-Year Treasury Note Futures); market indices such as the Dow Jones Industrial Index, NASDAQ Composite Index, Solar Energy Index, and Wind Energy Index; as well as the US Dollar Index. For each asset, the standard OHLCV (Open, High, Low, Close and Volume) market data are used. All data span from December 31, 2015, to November 14, 2024. Additionally, we have defined some sector-specific variables, to analyze the performance of our proposed solution approach under different conditions. These variables include Bond (CBOT2 and CBOT10), Green Energy (Solar Energy Index, Wind Energy Index and Natural Gas), Large Commodities (Coal, Copper, Gold, and Natural Gas) and Stocks (NASDAQ Composite Index and Dow Jones Industrial Index). For cross market experiments, the period from December 31, 2015 to November 30, 2023 is designated as the training period, and the period from December 1, 2023 to November 14, 2024, serves as the testing period.

In addition to market data from the WIND database, the study incorporates sentiment analysis using data from Twitter. This involves the collection of real-time tweets related to the markets or events, which are used to gauge public sentiment and opinions regarding various assets. Importantly, all Twitter data utilized in this study is authentically sourced from human users, including both individual leaders and key institutions, to ensure that the sentiment insights are based on genuine human interactions and expressions. There is no artificially generated data involved in this process, which helps maintain the integrity and authenticity of the sentiment analysis.

4.2. Computational efforts

All computational experiments were conducted using two NVIDIA RTX A5000 GPUs. The total computational cost for training the proposed RLDC-LLM was approximately 134 GPU hours, and the maximum memory usage was 48 GB.

Table 5

Performance evaluation of WTI trading strategies with various asset and sentiment/disagreement indicators.

States	Average return (%)	Sharpe ratio	Std.
S^{WTI}	2.82	0.47	0.25
$S^{WTI+Coal}$	20.16	0.73	0.25
$S^{WTI+Coal+D^b}$	16.62	0.69	0.21
$S^{WTI+Coal+D^c}$	40.83	2.17	0.18
$S^{WTI+Coal+D^b+D^c}$	27.69	1.29	0.20
$S^{WTI+Copper}$	3.19	0.05	0.26
$S^{WTI+Copper+D^b}$	23.02	1.66	0.13
$S^{WTI+Copper+D^c}$	13.91	0.46	0.26
$S^{WTI+Copper+D^b+D^c}$	23.71	0.97	0.22
$S^{WTI+C BOT2}$	11.3	1.1	0.08
$S^{WTI+C BOT2+D^b}$	16.98	0.88	0.17
$S^{WTI+C BOT2+D^c}$	9.34	0.38	0.19
$S^{WTI+C BOT2+D^b+D^c}$	20.42	1.03	0.18
$S^{WTI+C BOT10}$	10.19	0.69	0.12
$S^{WTI+C BOT10+D^b}$	8.47	0.41	0.16
$S^{WTI+C BOT10+D^c}$	28.16	1.32	0.20
$S^{WTI+C BOT10+D^b+D^c}$	38.39	1.95	0.19
$S^{WTI+DowJonesIndustrialindex}$	24.36	1.05	0.21
$S^{WTI+DowJonesIndustrialindex+D^b}$	31.75	1.35	0.22
$S^{WTI+DowJonesIndustrialindex+D^c}$	17.6	0.7	0.22
$S^{WTI+DowJonesIndustrialindex+D^b+D^c}$	33.47	1.5	0.21
$S^{WTI+Gold}$	19.77	1.16	0.15
$S^{WTI+Gold+D^b}$	34.21	1.39	0.23
$S^{WTI+Gold+D^c}$	43.96	2.66	0.16
$S^{WTI+Gold+D^b+D^c}$	20.97	0.77	0.25
$S^{WTI+Naturalgas}$	5.21	0.14	0.24
$S^{WTI+Naturalgas+D^b}$	20.32	1.01	0.18
$S^{WTI+Naturalgas+D^c}$	28.02	1.44	0.18
$S^{WTI+Naturalgas+D^b+D^c}$	9.13	0.31	0.23
$S^{WTI+NASDAQCompositeindex}$	19.76	0.74	0.24
$S^{WTI+NASDAQCompositeindex+D^b}$	19.67	1.09	0.16
$S^{WTI+NASDAQCompositeindex+D^c}$	20.42	1.02	0.18
$S^{WTI+NASDAQCompositeindex+D^b+D^c}$	44.83	2.3	0.19
$S^{WTI+SolarEnergyIndex}$	7.37	0.29	0.18
$S^{WTI+SolarEnergyIndex+D^b}$	19.08	1.01	0.17
$S^{WTI+SolarEnergyIndex+D^c}$	17.73	0.72	0.22
$S^{WTI+SolarEnergyIndex+D^b+D^c}$	15.59	0.54	0.25
$S^{WTI+USdollarindex}$	40.36	1.66	0.23
$S^{WTI+WindEnergyIndex}$	6.64	0.29	0.16
$S^{WTI+WindEnergyIndex+D^b}$	15.14	0.62	0.21
$S^{WTI+WindEnergyIndex+D^c}$	36.33	1.70	0.20
$S^{WTI+WindEnergyIndex+D^b+D^c}$	13.69	0.54	0.21

4.3. Cross market experiments

Table 5 presents the performance evaluation of West Texas Intermediate (WTI) crude oil trading strategies enhanced with various asset combinations and sentiment/disagreement indicators. The corresponding visualization results are shown in Fig. 8. The metrics assessed include Average Return (%), Sharpe Ratio, and Standard Deviation (Std.). The table systematically examines the impact of integrating different external assets (such as Coal, Copper, Gold, etc.) with WTI, alongside sentiment analysis and disagreement metrics (Basic Disagreement and Cross Disagreement).

Based on the results presented in Table 5, it is evident that strategies that incorporate an external asset with WTI consistently outperform the baseline strategy based solely on WTI, across all asset combinations. This underscores the value of diversifying the informational foundation beyond the crude oil market to enhance trading performance. It is also revealed the significant importance of incorporating LLMs into the crude oil futures trading strategy. Our experiments show that integrating LLMs with multiple assets and sentiment metrics consistently yields superior performance compared to traditional strategies, particularly when disagreement metrics and cross-market relationships are included. When we examine the results of asset combinations with

LLMs, certain combinations, such as WTI + Coal + Basic Disagreement and WTI + Copper + Basic Disagreement, exhibit significant improvements in average returns compared to their counterparts without disagreement metrics. Combinations like WTI + Coal + Cross Disagreement and WTI + NASDAQ Composite Index + Basic Disagreement + Cross Disagreement demonstrate substantial enhancements in performance metrics, indicating the efficacy of cross disagreement in capturing nuanced market dynamics. The most effective strategies, such as WTI + NASDAQ Composite index + Basic Disagreement + Cross Disagreement, achieve the highest average return, suggesting that the simultaneous integration of both basic and cross disagreement metrics alongside sentiment provides the robust enhancement to trading performance. When combined, Sentiment with Basic Disagreement and/or Cross Disagreement provides a comprehensive view of both the prevailing market sentiment and the underlying sources of opinion divergence. For example, in scenarios where market sentiment is generally optimistic but Cross Disagreement is high, this may indicate potential market instability or impending price volatility. This dual insight enables the model to capture more precise market signals, thereby enhancing its trading strategy.

A more detailed analysis reveals nuanced patterns when incorporating sentiment alongside disagreement metrics. While certain feature combinations demonstrate enhanced performance, the effectiveness

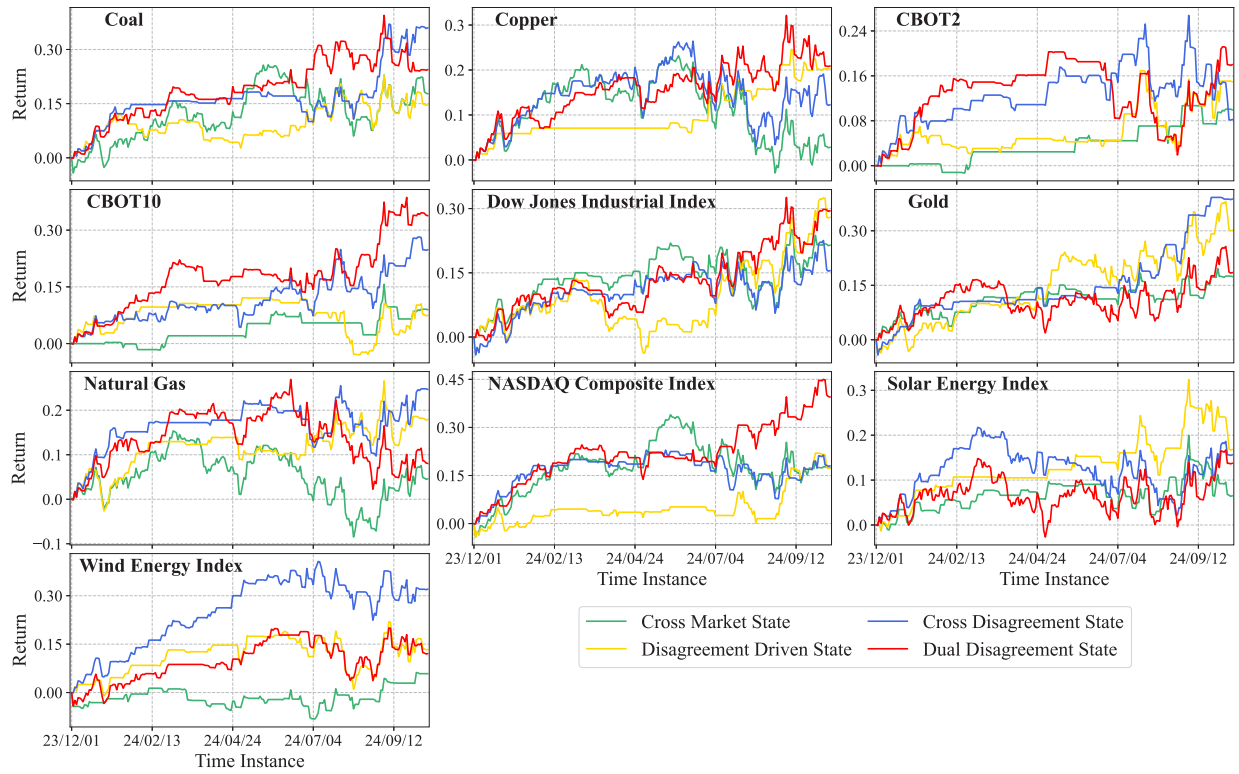


Fig. 8. Visualization of returns for different WTI trading strategies, Cross Market State, Disagreement Driven State, Cross Disagreement State, and Dual Disagreement State—across various assets, including coal, copper, gold, CBOT2, CBOT10, natural gas, NASDAQ composite index, solar energy index, wind energy index, and the Dow Jones Industrial Index.

of combining Basic and Cross Disagreement varies under different conditions:

- **Feature Redundancy:** Basic Disagreement and Cross Disagreement may provide similar information under certain market conditions. For instance, if the opinions of technical analysts and fundamental analysts consistently align, the overlap between Basic and Cross Disagreement increases, reducing the uniqueness of the combined feature set. This redundancy can cause the model to overemphasize certain features during training, neglecting other important signals and introducing noise that detracts from overall performance.
- **Market Contexts and Feature Effectiveness:** During volatile market phases, Cross Disagreement becomes particularly valuable as it captures heightened variance in opinions among diverse investor groups. For instance, technical analysts may focus on short-term price movements, while fundamental analysts prioritize long-term value. Cross Disagreement reflects these differing focal points, enhancing the model's ability to anticipate significant price fluctuations. In stable or trending market environments, Basic Disagreement is sufficient to capture essential opinion divergence within a homogeneous investor group. For example: Low Basic Disagreement indicates strong consensus supporting the current trend, allowing the model to follow the trend confidently. High Basic Disagreement signals potential trend reversals, providing critical information for timely trading decisions.

Table 6 demonstrates that the majority of asset combination strategies outperform baseline strategies that incorporate WTI with a single additional asset. The corresponding visualization results are shown in Fig. 9. Notably, integrating WTI with multiple assets, such as Large Commodities, consistently results in higher average returns compared to strategies based on WTI combined with a single asset (e.g., Copper or Gold alone). For instance, the strategy WTI + Large Commodities

achieves an impressive average return of 22.44%, markedly surpassing the 3.19% return of the single-asset WTI + Copper strategy. Similarly, the strategies WTI + Copper + Gold + Basic Disagreement and WTI + Large Commodities + Basic Disagreement + Cross Disagreement demonstrate substantial enhancements in performance metrics, with average returns of 25.37% and 37.92%, respectively.

These findings demonstrate that incorporating multiple assets into the strategy allows the RL model to capture a broader spectrum of market signals and investor behaviors, significantly enhancing its decision-making capabilities. For instance, Large Commodities, which include assets such as Coal, Copper, Gold, and Natural Gas, provide a holistic view of the commodities market and its interplay with crude oil prices. This diversification reduces the risks associated with relying solely on a single asset, thereby stabilizing returns and improving the overall robustness and resilience of the trading strategy.

While most multi-asset combinations outperform single-asset strategies, there are exceptions where multi-asset combinations fail to deliver superior returns. For example, The strategy WTI + Bond achieves an average return of 7.13%, lower than WTI + CBOT2 (11.3%) and WTI + CBOT10 (10.19%), despite the Bond sector being an aggregate of CBOT2 and CBOT10, likely due to the loss of unique signals from each individual component. The underperformance of the WTI + Bond strategy can be attributed to high correlation and multicollinearity between its constituent features, CBOT2 and CBOT10. When these two components are aggregated into a single Bond variable, the resulting feature may exhibit overlapping information, failing to provide additional unique insights. This redundancy can dilute the RL model's ability to effectively utilize the distinct signals from each Treasury Note Future, leading to diminished performance compared to strategies that incorporate CBOT2 and CBOT10 as separate features.

4.4. Comparison with baselines

To address the limitations of high correlation and multicollinearity, the study introduces the concept of connectedness in asset combination strategies, which has significantly improved trading performance.

Table 6

Performance metrics of WTI trading strategies with various combinations and sentiment/disagreement indicators.

States	Average return (%)	Sharpe ratio	Std.
$S^{WTI+Bond}$	7.13	0.23	0.22
$S^{WTI+Bond+D^b}$	21.1	0.98	0.19
$S^{WTI+Bond+D^c}$	10.37	0.63	0.13
$S^{WTI+Bond+D^b+D^c}$	22.06	1.08	0.19
$S^{WTI+Copper+Gold}$	18.65	1.10	0.15
$S^{WTI+Copper+Gold+D^b}$	25.37	2.79	0.08
$S^{WTI+Copper+Gold+D^c}$	27.38	2.01	0.13
$S^{WTI+Copper+Gold+D^b+D^c}$	27.67	1.3	0.20
$S^{WTI+GreenEnergy}$	13.49	0.62	0.19
$S^{WTI+GreenEnergy+D^b}$	13.06	0.45	0.25
$S^{WTI+GreenEnergy+D^c}$	27.53	1.38	0.18
$S^{WTI+GreenEnergy+D^b+D^c}$	13.25	0.89	0.13
$S^{WTI+LargeCommodities}$	22.44	0.91	0.22
$S^{WTI+LargeCommodities+D^b}$	16.46	0.62	0.23
$S^{WTI+LargeCommodities+D^c}$	25.65	1.65	0.14
$S^{WTI+LargeCommodities+D^b+D^c}$	37.92	1.93	0.19
$S^{WTI+Stocks}$	19.25	0.68	0.25
$S^{WTI+Stocks+D^b}$	16.63	0.78	0.19
$S^{WTI+Stocks+D^c}$	18.22	0.71	0.23
$S^{WTI+Stocks+D^b+D^c}$	23.7	1.57	0.14

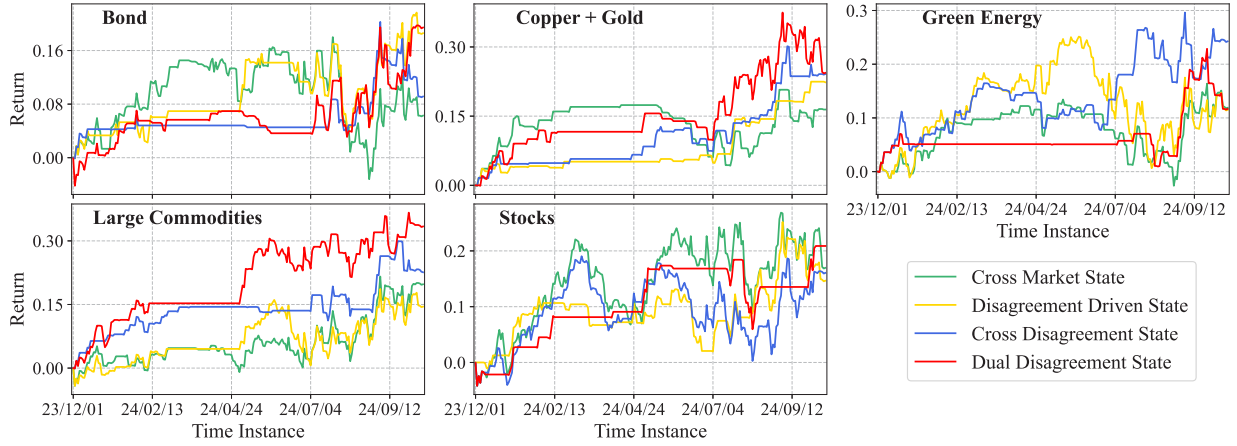


Fig. 9. Visualization of returns for various WTI trading strategies, Cross Market State, Disagreement Driven State, Cross Disagreement State, and Dual Disagreement State, across multiple sectors, including bonds, copper + gold, green energy, large commodities, and stocks.

Table 7

Performance comparison between predication based-methods [102] with the RLDC-LLM-based methods.

Compare methods	Average return (%)	Sharpe ratio	Std.
SARIMA-X [102]	27.54	1.51	0.18
SVR-X [102]	34.11	1.75	0.20
ELM-X [102]	36.89	1.62	0.23
BPNN-X [102]	40.29	1.76	0.23
LSTM-X [102]	42.23	1.70	0.25
Informer-X [102]	50.37	1.88	0.27
RLDC-LLM (Cross Market State)	59.12	1.67	0.34
RLDC-LLM (Disagreement Driven State)	80.35	2.30	0.34
RLDC-LLM (Cross Disagreement State)	74.62	1.69	0.43
RLDC-LLM (Dual Disagreement State)	86.92	1.91	0.44
RLDC-LLM (Connectedness-based State)	105.51	3.20	0.32

Connectedness ensures that selected assets are not only diverse but also meaningfully interconnected, providing the RL model with a richer and more nuanced understanding of market dynamics. Specifically, the RLDC-LLM-based methods leverage both static connectedness and dynamic connectedness, as shown in Table 7, to improve their ability to capture both stable and evolving relationships between assets.

In order to have a fair comparison with the baseline methods presented in [102], the training and testing periods for our proposed

RLDC-LLM has been adjust accordingly. Table 7 demonstrates the superior performance of the RLDC-LLM framework compared to traditional time-series models, machine learning approaches, and deep learning-based methods across all evaluation metrics, including average return, Sharpe ratio, and standard deviation. Among the RLDC-LLM configurations, the Connectedness-based State achieves the highest average return (105.51%), Sharpe ratio (3.20), and maintains a low standard deviation (0.32), underscoring its ability to effectively balance risk and

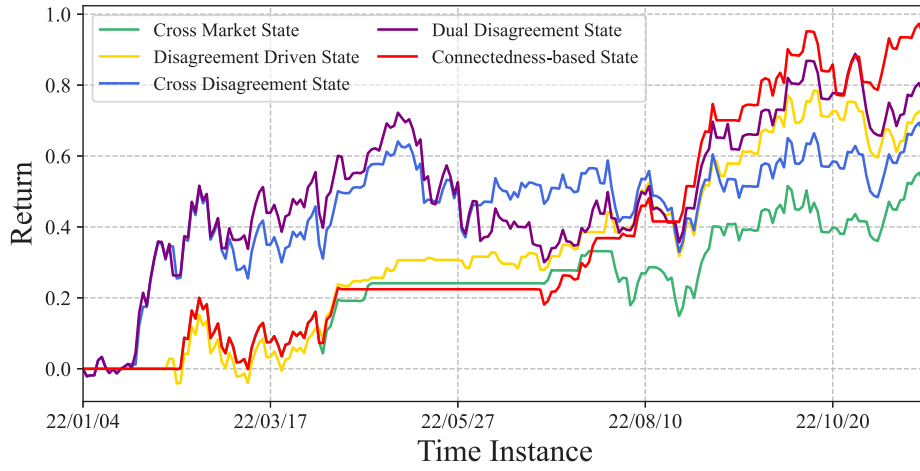


Fig. 10. Visualization of the performance of RLDC-LLM configurations, including Cross Market State, Disagreement Driven State, Cross Disagreement State, Dual Disagreement State and connectedness-based state.

return. The Disagreement Driven State, Cross Disagreement State and Dual Disagreement State also outperform competing methods, further validating the efficacy of incorporating disagreement and connectedness metrics in enhancing trading strategies. In contrast, traditional methods like SARIMA-X and machine learning models such as SVR-X and ELM-X exhibit lower average returns and Sharpe ratios, reflecting their limited capacity to capture the complex interdependencies in financial markets. Deep learning approaches, particularly Informer-X, perform better than traditional methods, with notable improvements in average returns (50.37%) and Sharpe ratio (1.88). However, the RLDC-LLM framework demonstrates its superiority by leveraging advanced market features, such as disagreement metrics and connectedness dynamics, enabling it to outperform all baseline models. This analysis highlights the robustness and adaptability of RLDC-LLM in dynamic financial environments, making it a highly effective framework for crude oil trading strategies. The corresponding visualization results of the RLDC-LLM configurations are illustrated in Fig. 10. The figure highlights the performance differences across these configurations, showcasing the effectiveness of incorporating disagreement metrics and connectedness in enhancing trading strategies.

4.5. Effectiveness of forgetting mechanism under different market conditions

To evaluate the effectiveness of the proposed forgetting mechanism across varying market conditions, we conducted experiments within both stable and volatile environments. Our study involved a comparative analysis of two distinct groups of models: one employing a uniform decay rate, fixed at $\Theta = 3.84$ across all scenarios, and another utilizing a dynamic decay rate, set at $\Theta = 3.84$ during stable periods and accelerated to $\Theta = 1.92$ during volatile periods. We identified two historical intervals as stable (December 14, 2017–December 10, 2019, and March 01, 2024–November 14, 2024) and two as volatile (December 31, 2015–December 13, 2017, and December 11, 2019–February 29, 2024). Each variant of the model underwent three trials with distinct random seeds, and we report the average performance across these runs. The results, illustrated in Fig. 11, clearly demonstrate that the uniform-rate model outperformed the dynamic-rate model in both volatile environments (from December 01, 2023 to February 29, 2024) and stable environments (from March 01, 2024 to November 14, 2024). In volatile conditions, the model with a uniform decay rate exhibited a superior return growth rate compared to its dynamic-rate counterpart. As volatility diminished, the uniform model's returns remained approximately 10% higher than those of the dynamic model. Although the performance advantage of the uniform-rate model in stable environments was less pronounced than in volatile conditions,

it consistently maintained superiority, outperforming the dynamic-rate model by approximately 5%.

We hypothesized that an adaptive forgetting rate would better handle regime shifts by accelerating memory decay in volatile environments. However, our results demonstrated that a fixed Θ outperformed the regime-dependent approach. This likely stems from the stability provided by a uniform exponential decay rate, which consistently smooths sentiment signals, filtering out high-frequency noise and yielding robust indicators. In contrast, dynamically adjusting the forgetting rate can lead to overfitting transient fluctuations, rather than capturing meaningful long-term trends. This reflects the classic bias–variance trade-off: while an adaptive rate might better fit different market regimes, it also heightens sensitivity to random noise, undermining robustness. Empirical studies in time-series forecasting support this tendency, showing that overly flexible decay mechanisms often produce unstable predictions [103]. A fixed Θ acts as a stabilizing filter, smoothing short-term volatility while preserving the underlying market sentiment trajectory.

Moreover, a single decay rate aligns with behavioral patterns in financial markets. Human attention and memory typically follow an exponential decay curve, evident in recency bias and anchoring effects. These biases suggest that investors process information through gradual forgetting. A fixed Θ captures this “average” memory span, reflecting how news and events are digested over time, aligning the model with natural investor information integration. Empirical evidence from market microstructure further supports this view. Studies on news impact half-lives [104] identify two decay components: a fast-reverting component with a half-life of about one day, and a slower component with a half-life around 25 days. High-frequency sentiment shocks often revert quickly, contributing more to noise than lasting price changes, while the slower component aligns with sustained price movements. A uniform Θ emphasizes the slower, persistent sentiment component by filtering out ephemeral noise, focusing on enduring signals and avoiding distractions from short-lived fluctuations. This fixed-rate approach approximates the typical timescale of investor memory, preventing distortion from over-parameterization.

Additionally, the dynamic-rate approach faces practical challenges: reliably identifying stable versus volatile markets is non-trivial. Market regime classification is subjective and lacks a universal standard, with gradual volatility changes and ambiguous boundaries between “stable” and “volatile” periods. Misclassification can lead to inappropriate decay rates, degrading performance. Thus, the fixed-rate approach’s robustness is highlighted, as it mitigates noise, aligns with investor behavior, and avoids risks of regime misclassification, ensuring reliable performance across varied market conditions.

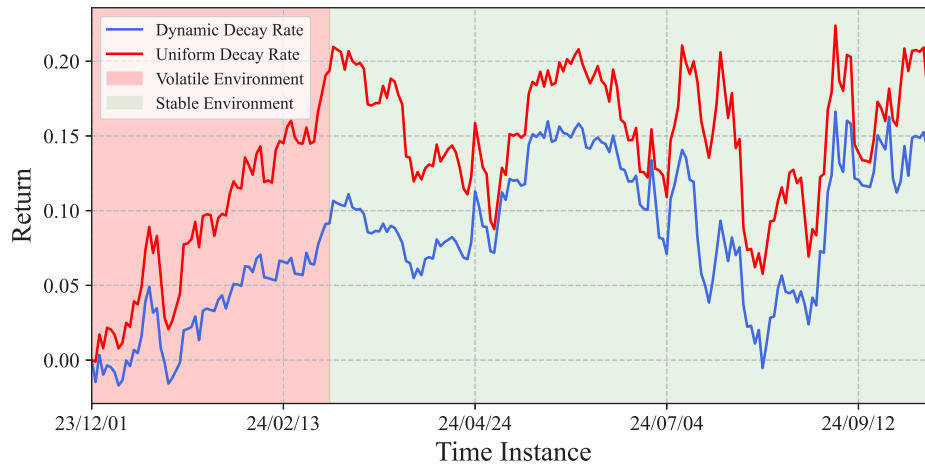


Fig. 11. Visualization of the performance of the RLDC-LLM model across various market conditions, considering different decay rates in the forgetting mechanism.

Table 8

Model performance comparison.

Metric	Method	Mean	95% CI	Statistical significance
Average Return (%)	RLDC-LLM (Connectedness-based State)	105.51	[101.93, 109.09]	Yes ($p < 0.05$)
	RLDC-LLM (Dual Disagreement State)	86.92	[83.34, 90.50]	Yes ($p < 0.05$)
	Benchmark Model: Informer-X	50.37	[48.22, 52.52]	–
Sharpe Ratio	RLDC-LLM (Connectedness-based State)	3.20	[2.94, 3.43]	Yes ($p < 0.05$)
	RLDC-LLM (Dual Disagreement State)	1.91	[1.77, 2.05]	Yes ($p < 0.05$)
	Benchmark Model: Informer-X	1.88	[1.74, 2.02]	–

4.6. Model robustness

To ensure the robustness of our findings, we conducted comprehensive statistical evaluations incorporating multiple validation techniques. For both cumulative returns and Sharpe ratios, we calculated 95% confidence intervals using t-distributions suitable for small sample sizes ($n = 10$), with the results summarized in Table 8. Regarding average returns, the RLDC-LLM model with a Connectedness-based State achieved a mean return of 105.51% (95% CI: [101.93%, 109.09%]), significantly outperforming the Benchmark Informer-X model's 50.37% (95% CI: [48.22%, 52.52%]). Similarly, the Dual Disagreement State configuration yielded an average return of 86.92% (95% CI: [83.34%, 90.50%]), demonstrating consistent superiority across state representations.

In terms of risk-adjusted performance, the Sharpe ratios exhibited similar patterns. Our analyses indicate that while RLDC-LLM variants achieved Sharpe ratios slightly exceeding that of the Benchmark Informer-X (1.88 versus 1.91 and 3.20), their true advantage lies in significantly higher absolute returns. For example, the Connectedness-based State achieved a mean return of 105.51%, more than double the Benchmark's 50.37%, despite only marginally higher risk-adjusted metrics. All comparisons were statistically significant at the $p < 0.05$ level, as confirmed by paired t-tests, with non-overlapping confidence intervals further validating these differences.

To address potential temporal dependencies and market regime shifts, we employed bootstrapping and rolling-window analyses, which confirmed stable performance across varying timeframes and conditions. Variance decomposition affirmed the explanatory power of disagreement and connectedness features, underscoring their critical role in capturing asset return dynamics. Collectively, these analyses substantiate the systematic advantages of RLDC-LLM over traditional benchmarks.

5. Conclusion

This study introduces RLDC-LLM (Reinforcement Learning Incorporating Disagreement and Connectedness Methods with Large Language

Models), a novel framework designed to optimize trading strategies in the WTI crude oil futures market. By integrating reinforcement learning with large language models, RLDC-LLM effectively incorporates underutilized yet critical factors such as macroeconomic influences, green energy trends, and investor sentiment, reflecting the increasing complexity and interconnectedness of the modern energy market. Through the combination of structured time-series data and unstructured textual data, the framework provides a holistic approach to capturing diverse market signals, offering valuable insights into the dynamics of a rapidly evolving energy landscape.

The RLDC-LLM framework systematically integrates four distinct state configurations—Cross Market State, Disagreement Driven State, Cross Disagreement State, and Dual Disagreement State—alongside static and dynamic connectedness metrics. These components work synergistically to capture sentiment, disagreement, and cross-market dynamics while modeling the stable and time-varying interdependencies among market variables. The inclusion of a chain-of-reasoning prompt technique further enhances the framework by classifying investor types, distinguishing between sentiment-driven disagreement and cross-disagreement, and providing a nuanced understanding of market behaviors. Experimental results demonstrate the framework's superior performance across key metrics, with the Dual Disagreement State achieving the highest average return, Sharpe ratio, and lowest volatility. These findings underscore RLDC-LLM's ability to adapt to fluctuating market conditions and deliver robust trading performance, outperforming traditional price-prediction-based strategies.

Beyond the technical contributions, RLDC-LLM addresses broader challenges faced by the energy market, particularly in the context of the global transition toward decarbonization. The energy market is undergoing a profound transformation as nations and industries strive to reduce carbon emissions and integrate renewable energy sources. This shift introduces new complexities, such as the interplay between traditional fossil fuels and emerging green energy technologies, as well as the heightened influence of investor sentiment and macroeconomic trends. Energy brokers, as key intermediaries in this transition, must

navigate these complexities by leveraging advanced tools and strategies to manage risks, optimize trading, and align with sustainability objectives. RLDC-LLM offers a powerful framework for understanding and responding to these challenges, providing a pathway for brokers and other market participants to thrive in this dynamic environment. By addressing the challenges of integrating heterogeneous data sources, modeling interconnected market dynamics, and analyzing investor sentiment and disagreement, the framework provides a comprehensive tool for navigating the complexities of the modern energy landscape. As the global energy market continues its transition toward decarbonization, frameworks like RLDC-LLM will play a critical role in equipping energy brokers and other market participants with the tools needed to adapt, innovate, and succeed in this transformative era.

One important avenue for future research lies in expanding the scope of external asset combinations and incorporating emerging indicators that reflect the ongoing transition to a low-carbon economy. While this study focuses on a fixed set of assets, future studies could include metrics such as ESG (Environmental, Social, and Governance) ratings, renewable energy benchmarks, or carbon credit prices to better align with the evolving priorities of the energy market. Additionally, integrating advanced hierarchical models could enhance the RLDC-LLM framework's ability to capture multi-scale dependencies across different market dimensions, further improving its robustness and predictive power.

CRedit authorship contribution statement

Tianxiang Cui: Writing – review & editing, Writing – original draft, Validation, Supervision, Methodology, Formal analysis, Conceptualization. **Yujian Ye:** Validation, Investigation, Funding acquisition, Conceptualization. **Yiran Li:** Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation. **Nanjiang Du:** Visualization, Validation. **Xingke Song:** Visualization, Validation. **Yicheng Zhu:** Visualization, Validation. **Xiaoying Yang:** Visualization, Validation. **Goran Strbac:** Writing – review & editing, Resources, Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work was supported by the National Natural Science Foundation of China (52477083, 52207082), Natural Science Foundation of Jiangsu Province, China (BK20220842), Ningbo Natural Science Foundation, China (2023J194), University of Nottingham Ningbo China Education Foundation (LDS202303) and Basic and Commonweal Programme of Zhejiang Natural Science Foundation, China (LY24F020006).

Data availability

Data will be made available on request.

References

- [1] Jia Zhijie, Wen Shiyan, Lin Boqiang. The effects and reacts of COVID-19 pandemic and international oil price on energy, economy, and environment in China. *Appl Energy* 2021;302:117612.
- [2] May Daniel C, Taylor Matthew, Musilek Petr. Decentralized coordination of distributed energy resources through local energy markets and deep reinforcement learning. *Energy AI* 2024;18:100446.
- [3] Mu Yunfei, Guo Haochen, Wu Zhijun, Jia Hongjie, Jin Xiaolong, Qi Yan. A two-layer low-carbon economic planning method for park-level integrated energy systems with carbon-energy synergistic hub. *Energy AI* 2024;18:100435.
- [4] Demir Sumeyra, Stappers Bart, Kok Koen, Paterakis Nikolaos G. Statistical arbitrage trading on the intraday market using the asynchronous advantage actor-critic method. *Appl Energy* 2022;314:118912.
- [5] Pan Mingzhang, Cao Xinxin, Fu Changcheng, Liao Shengyou, Zhou Xiaorong, Guan Wei. Emission prediction and optimization of methanol/diesel dual-fuel engines based on itransformer-bigru and NSGA-III. *Energy AI* 2025;19:100466.
- [6] International Energy Agency. *World Energy Outlook 2024*. International Energy Agency; 2024.
- [7] Zhang Qingjun, Wei Renyi, Fan Sijia. New energy mineral price shocks and volatility responses in green securities markets: Structural effects and dynamic spillovers. *Appl Energy* 2025;382:125230.
- [8] Han Te, Cong Rong-Gang, Yu Biying, Tang Baojun, Wei Yi-Ming. Integrating local knowledge with ChatGPT-like large-scale language models for enhanced societal comprehension of carbon neutrality. *Energy AI* 2024;18:100440.
- [9] Xing Frank Z, Cambria Erik, Zhang Yue. Sentiment-aware volatility forecasting. *Knowl-Based Syst* 2019;176:68–76.
- [10] Gao George P, Lu Xiaomeng, Song Zhaogang, Yan Hongjun. Disagreement beta. *J Monet Econ* 2019;107:96–113.
- [11] Ogbuabor Jonathan E, Ukwueze Ezebuilo R, Mba Ifeoma C, Ojonta Obed I, Orji Anthony. The asymmetric impact of economic policy uncertainty on global retail energy markets: Are the markets responding to the fear of the unknown? *Appl Energy* 2023;334:120671.
- [12] Koroteev Dmitry, Tekic Zeljko. Artificial intelligence in oil and gas upstream: Trends, challenges, and scenarios for the future. *Energy AI* 2021;3:100041.
- [13] Yavari Mohammad, Bohreghi Iman Mohammadi. Developing a green-resilient power network and supply chain: Integrating renewable and traditional energy sources in the face of disruptions. *Appl Energy* 2025;377:124654.
- [14] Zhang Steven, May Daniel, Gül Mustafa, Musilek Petr. Reinforcement learning-driven local transactive energy market for distributed energy resources. *Energy AI* 2022;8:100150.
- [15] Brown Tom, Mann Benjamin, Ryder Nick, Subbiah Melanie, Kaplan Jared D, Dhariwal Prafulla, et al. Language models are few-shot learners. *Adv Neural Inf Process Syst* 2020;33:1877–901.
- [16] Wei Jason, Wang Xuezhai, Schuurmans Dale, Bosma Maarten, Xia Fei, Chi Ed, et al. Chain-of-thought prompting elicits reasoning in large language models. *Adv Neural Inf Process Syst* 2022;35:24824–37.
- [17] Cheng Junyan, Chin Peter. Sociodojo: Building lifelong analytical agents with real-world text and time series. In: *The twelfth international conference on learning representations*. 2024.
- [18] Kirtac Kemal, Germano Guido. Sentiment trading with large language models. *Financ Res Lett* 2024;62:105227.
- [19] Fatourous Georgios, Metaxas Konstantinos, Soldatos John, Kyriazis Dimosthenis. Can large language models beat wall street? unveiling the potential of ai in stock selection. 2024, arXiv preprint arXiv:2401.03737.
- [20] Xing Frank. Designing heterogeneous llm agents for financial sentiment analysis. *ACM Trans Manag Inf Syst* 2024.
- [21] Koa Kelvin JL, Ma Yunshan, Ng Ritchie, Chua Tat-Seng. Learning to generate explainable stock predictions using self-reflective large language models. In: *Proceedings of the ACM on web conference 2024*. 2024, p. 4304–15.
- [22] Schulman John, Wolski Filip, Dhariwal Prafulla, Radford Alec, Klimov Oleg. Proximal policy optimization algorithms. 2017, arXiv preprint arXiv:1707.06347.
- [23] Liu Tangyong, Gong Xu. Analyzing time-varying volatility spillovers between the crude oil markets using a new method. *Energy Econ* 2020;87:104711.
- [24] Cui Tianxiang, Du Nanjiang, Yang Xiaoying, Ding Shusheng. Multi-period portfolio optimization using a deep reinforcement learning hyper-heuristic approach. *Technol Forecast Soc Change* 2024;198:122944.
- [25] Jadidzadeh Ali, Serletis Apostolos. How does the US natural gas market react to demand and supply shocks in the crude oil market? *Energy Econ* 2017;63:66–74.
- [26] Liu Jing, Ma Feng, Tang Yingkai, Zhang Yaojie. Geopolitical risk and oil volatility: A new insight. *Energy Econ* 2019;84:104548.
- [27] Ding Shusheng, Wang Kaihao, Cui Tianxiang, Du Min. The time-varying impact of geopolitical risk on natural resource prices: The post-COVID era evidence. *Resour Policy* 2023;86:104161.
- [28] Bali Turan G, Brown Stephen J, Tang Yi. Is economic uncertainty priced in the cross-section of stock returns? *J Financ Econ* 2017;126(3):471–89.
- [29] Yang Cai, Gong Xu, Zhang Hongwei. Volatility forecasting of crude oil futures: The role of investor sentiment and leverage effect. *Resour Policy* 2019;61:548–63.
- [30] Batten Jonathan A, Brzezczynski Janusz, Ciner Cetin, Lau Marco CK, Lucey Brian, Yarovaya Larisa. Price and volatility spillovers across the international steam coal market. *Energy Econ* 2019;77:119–38.
- [31] Li Jianglong, Xie Chunping, Long Houyin. The roles of inter-fuel substitution and inter-market contagion in driving energy prices: Evidences from China's coal market. *Energy Econ* 2019;84:104525.
- [32] Mensi Walid, Rehman Mobeen Ur, Maitra Debasish, Al-Yahyaee Khamis Hamed, Vo Xuan Vinh. Oil, natural gas and BRICS stock markets: Evidence of systemic risks and co-movements in the time-frequency domain. *Resour Policy* 2021;72:102062.

- [33] Ding Lili, Zhao Zhongchao, Wang Lei. Probability density forecasts for natural gas demand in China: Do mixed-frequency dynamic factors matter? *Appl Energy* 2022;312:118756.
- [34] Bergek Anna, Berggren Christian, Magnusson Thomas, Hobday Michael. Technological discontinuities and the challenge for incumbent firms: Destruction, disruption or creative accumulation? *Res Policy* 2013;42(6–7):1210–24.
- [35] Mäkitie Tuukka, Andersen Allan D, Hanson Jens, Normann Håkon E, Thune Taran M. Established sectors expediting clean technology industries? The norwegian oil and gas sector's influence on offshore wind power. *J Clean Prod* 2018;177:813–23.
- [36] Li Jingpeng, Umar Muhammad, Huo Jiale. The spillover effect between Chinese crude oil futures market and Chinese green energy stock market. *Energy Econ* 2023;119:106568.
- [37] Geng Jiang-Bo, Liu Changyu, Ji Qiang, Zhang Dayong. Do oil price changes really matter for clean energy returns? *Renew Sustain Energy Rev* 2021;150:111429.
- [38] Lu Quanying, Li Yuze, Chai Jian, Wang Shouyang. Crude oil price analysis and forecasting: A perspective of “new triangle”. *Energy Econ* 2020;87:104721.
- [39] Zhang Yue-Jun, Wang Zi-Yi. Investigating the price discovery and risk transfer functions in the crude oil and gasoline futures markets: Some empirical evidence. *Appl Energy* 2013;104:220–8.
- [40] Ding Shusheng, Zheng Dandan, Cui Tianxiang, Du Min. The oil price-inflation nexus: The exchange rate pass-through effect. *Energy Econ* 2023;125:106828.
- [41] Tiwari Aviral Kumar, Mishra Bibhuti Ranjan, Solarin Sakiru Adebola. Analysing the spillovers between crude oil prices, stock prices and metal prices: The importance of frequency domain in USA. *Energy* 2021;220:119732.
- [42] Coronado Semei, Jimnez-Rodriguez Rebeca, Rojas Omar. An empirical analysis of the relationships between crude oil, gold and stock markets. *Energy J* 2018;39(1_suppl):193–208.
- [43] Wang Lu, Ruan Hang, Hong Yanran, Luo Keyu. Detecting the hidden asymmetric relationship between crude oil and the US dollar: A novel neural granger causality method. *Res Int Bus Financ* 2023;64:101899.
- [44] Xiao Jihong, Hu Chunyan, Ouyang Guangda, Wen Fenghua. Impacts of oil implied volatility shocks on stock implied volatility in China: Empirical evidence from a quantile regression approach. *Energy Econ* 2019;80:297–309.
- [45] Degiannakis Stavros, Filis George, Arora Vipin. Oil prices and stock markets: A review of the theory and empirical evidence. *Energy J* 2018;39(5):85–130.
- [46] Efimova Olga, Serletis Apostolos. Energy markets volatility modelling using GARCH. *Energy Econ* 2014;43:264–73.
- [47] Chen Yan, Qiao Gaoxiu, Zhang Feipeng. Oil price volatility forecasting: Threshold effect from stock market volatility. *Technol Forecast Soc Change* 2022;180:121704.
- [48] Wen Fenghua, Zhao Yupei, Zhang Minzhi, Hu Chunyan. Forecasting realized volatility of crude oil futures with equity market uncertainty. *Appl Econ* 2019;51(59):6411–27.
- [49] Li Lei, Yin Libo, Zhou Yimin. Exogenous shocks and the spillover effects between uncertainty and oil price. *Energy Econ* 2016;54:224–34.
- [50] Bašta Milan, Molnár Peter. Oil market volatility and stock market volatility. *Financ Res Lett* 2018;26:204–14.
- [51] Gargano Antonio, Pettenuzzo Davide, Timmermann Allan. Bond return predictability: Economic value and links to the macroeconomy. *Manag Sci* 2019;65(2):508–40.
- [52] Kang Wensheng, Ratti Ronald A, Yoon Kyung Hwan. The impact of oil price shocks on US bond market returns. *Energy Econ* 2014;44:248–58.
- [53] Morrison Eleanor J. Energy price implications for emerging market bond returns. *Res Int Bus Financ* 2019;50:398–415.
- [54] Dai Zhifeng, Kang Jie. Bond yield and crude oil prices predictability. *Energy Econ* 2021;97:105205.
- [55] Xiang Ying, Zhuang Xiao Hong. Application of ARIMA model in short-term prediction of international crude oil price. *Adv Mater Res* 2013;798:979–82.
- [56] Lindberg Oskar, Lingfors David, Arnqvist Johan, van Der Meer Dennis, Munkhammar Joakim. Day-ahead probabilistic forecasting at a co-located wind and solar power park in Sweden: Trading and forecast verification. *Adv Appl Energy* 2023;9:100120.
- [57] Kilian Lutz. Not all oil price shocks are alike: Disentangling demand and supply shocks in the crude oil market. *Am Econ Rev* 2009;99(3):1053–69.
- [58] Chen Engu, He Xin James. Crude oil price prediction with decision tree based regression approach. *J Int Technol Inf Manag* 2019;27(4):2–16.
- [59] Movagharnjad Kamyar, Mehdizadeh Bahman, Banihashemi Morteza, Kordkheili Masoud Sheikhhi. Forecasting the differences between various commercial oil prices in the Persian gulf region by neural network. *Energy* 2011;36(7):3979–84.
- [60] Gupta Varun, Pandey Ankit. Crude oil price prediction using LSTM networks. *Int J Comput Inf Eng* 2018;12(3):226–30.
- [61] Sun Shaolong, Sun Yuying, Wang Shouyang, Wei Yunjie. Interval decomposition ensemble approach for crude oil price forecasting. *Energy Econ* 2018;76:274–87.
- [62] Gruver Nate, Finzi Marc, Qiu Shikai, Wilson Andrew G. Large language models are zero-shot time series forecasters. *Adv Neural Inf Process Syst* 2023;36:19622–35.
- [63] Xue Hongkuan, Li Jian, Zhang Tianlin. TimeBERT: Pre-training language models with temporal reasoning. *Indian Competit Law Rev* 2023.
- [64] Liu Yong, Zeng Haixu, Cao Ming. iTransformer: Inverted transformers are effective for time series forecasting. In: *NeurIPS*. 2023.
- [65] Liu Peiyuan, Guo Hang, Dai Tao, Li Naiqi, Bao Jigang, Ren Xudong, et al. Calf: Aligning llms for time series forecasting via cross-modal fine-tuning. In: *Proceedings of the AAAI conference on artificial intelligence*, vol. 39, (18):2025, p. 18915–23.
- [66] Liu Chenxi, Yang Sun, Xu Qianxiong, Li Zhishuai, Long Cheng, Li Ziyue, et al. Spatial-temporal large language model for traffic prediction. In: *2024 25th IEEE international conference on mobile data management*. IEEE; 2024, p. 31–40.
- [67] Zhou Tian, Niu Peisong, Sun Liang, Jin Rong, et al. One fits all: Power general time series analysis by pretrained lm. *Adv Neural Inf Process Syst* 2023;36:43322–55.
- [68] Jin Ming, Wang Shiyu, Ma Lintao, Chu Zhixuan, Zhang James Y, Shi Xiaoming, et al. Time-llm: Time series forecasting by reprogramming large language models. 2023, arXiv preprint arXiv:2310.01728.
- [69] Liu Yong, Qin Guo, Huang Xiangdong, Wang Jianmin, Long Mingsheng. Autotimes: Autoregressive time series forecasters via large language models. *Adv Neural Inf Process Syst* 2024;37:122154–84.
- [70] Yu Xinli, Chen Zheng, Ling Yuan, Dong Shujing, Liu Zongyi, Lu Yanbin. Temporal data meets LLM—explainable financial time series forecasting. 2023, arXiv preprint arXiv:2306.11025.
- [71] Wang Meiyun, Izumi Kiyoshi, Sakaji Hiroki. LLMFactor: Extracting profitable factors through prompts for explainable stock movement prediction. 2024, arXiv preprint arXiv:2406.10811.
- [72] Tan Mingtian, Merrill Mike, Gupta Vinayak, Althoff Tim, Hartvigsen Tom. Are language models actually useful for time series forecasting? *Adv Neural Inf Process Syst* 2024;37:60162–91.
- [73] Du Nanjiang, Zhao Yida, Wang Jintao, Zhu Yicheng, Xie Siyu, Yang Luyao, et al. Enhancing cryptocurrency trading strategies: A deep reinforcement learning approach integrating multi-source LLM sentiment analysis. In: *2025 IEEE symposium series on computational intelligence*. IEEE; 2025.
- [74] Antweiler Werner, Frank Murray Z. Is all that talk just noise? The information content of internet stock message boards. *J Financ* 2004;59(3):1259–94.
- [75] Cookson J Anthony, Niessner Marina. Why don't we agree? Evidence from a social network of investors. *J Financ* 2020;75(1):173–228.
- [76] Diebold Francis X, Yilmaz Kamil. On the network topology of variance decompositions: Measuring the connectedness of financial firms. *J Econometrics* 2014;182(1):119–34.
- [77] za Demirer Ri, Ferrer Román, Shahzad Syed Jawad Hussain. Oil price shocks, global financial markets and their connectedness. *Energy Econ* 2020;88:104771.
- [78] Kaul Byrne. Effects of crude oil prices on copper and maize prices. *Futur Bus J* 2021;7(1):54.
- [79] Ahmadi Maryam, Behmiri Niaz Bashiri, Manera Matteo. How is volatility in commodity markets linked to oil price shocks? *Energy Econ* 2016;59:11–23.
- [80] Choudhury Tonmoy, Kinatader Harald, Neupane Biwesh. Gold, bonds, and epidemics: A safe haven study. *Financ Res Lett* 2022;48:102978.
- [81] Lee Yongjae, Kim Woo Chang, Kim Jang Ho. Achieving portfolio diversification for individuals with low financial sustainability. *Sustainability* 2020;12(17):7073.
- [82] Górka Joanna, Kuziak Katarzyna. Dynamic connectedness among alternative and conventional energy etfs based on the TVP-var approach. *Energies* 2024;17(23):5929.
- [83] Vatcheva Kristina P, Lee MinJae. Multicollinearity in regression analyses conducted in epidemiologic studies. *Epidemiology* 2016;6(2):227.
- [84] BenSaïda Ahmed. Good and bad volatility spillovers: An asymmetric connectedness. *J Financ Mark* 2019;43:78–95.
- [85] Naeem Muhammad Abubakr, Balli Faruk, Shahzad Syed Jawad Hussain, de Bruin Anne. Energy commodity uncertainties and the systematic risk of US industries. *Energy Econ* 2020;85:104589.
- [86] Ferrer Román, Shahzad Syed Jawad Hussain, López Raquel, Jareño Francisco. Time and frequency dynamics of connectedness between renewable energy stocks and crude oil prices. *Energy Econ* 2018;76:1–20.
- [87] Zhang Yue-Jun, Fan Ying, Tsai Hsien-Tang, Wei Yi-Ming. Spillover effect of US dollar exchange rate on oil prices. *J Policy Model* 2008;30(6):973–91.
- [88] Shang Jin, Hamori Shigeyuki. Quantile time-frequency connectedness analysis between crude oil, gold, financial markets, and macroeconomic indicators: Evidence from the US and EU. *Energy Econ* 2024;132:107473.
- [89] Mensi Walid, Aslan Aylin, Vo Xuan Vinh, Kang Sang Hoon. Time-frequency spillovers and connectedness between precious metals, oil futures and financial markets: Hedge and safe haven implications. *Int Rev Econ Financ* 2023;83:219–32.
- [90] Asadi Mehrad, Roubaud David, Tiwari Aviral Kumar. Volatility spillovers amid crude oil, natural gas, coal, stock, and currency markets in the US and China based on time and frequency domain connectedness. *Energy Econ* 2022;109:105961.
- [91] Guo Yanfeng, Zhao HuanYu. Volatility spillovers between oil and coal prices and its implications for energy portfolio management in China. *Int Rev Econ Financ* 2024;89:446–57.

- [92] Perifanis Theodosios, Dagoumas Athanasios. Price and volatility spillovers between the US crude oil and natural gas wholesale markets. *Energies* 2018;11(10):2757.
- [93] Alharbey Mohammed, Alfahaid Turki Mohammed, Ben-Salha Ousama. Asymmetric volatility spillover between oil prices and regional renewable energy stock markets: A time-varying parameter vector autoregressive-based connectedness approach. *AIMS Math* 2023;8:30639–67.
- [94] Antonakakis Nikolaos, Chatziantoniou Ioannis, Gabauer David. Refined measures of dynamic connectedness based on time-varying parameter vector autoregressions. *J Risk Financ Manag* 2020;13(4):84.
- [95] Lütkepohl Helmut. New introduction to multiple time series analysis. Springer Science & Business Media; 2005.
- [96] Cui Tianxiang, Yang Xiaoying, Jia Fuhua, Jin Jiahuan, Ye Yujian, Bai Ruibin. Mobile robot sequential decision making using a deep reinforcement learning hyper-heuristic approach. *Expert Syst Appl* 2024;257:124959.
- [97] Jin Jiahuan, Cui Tianxiang, Bai Ruibin, Qu Rong. Container port truck dispatching optimization using Real2Sim based deep reinforcement learning. *European J Oper Res* 2024;315(1):161–75.
- [98] Yang Xiaoying, Cui Tianxiang, Wang Hongru, Ye Yujian. Multiagent deep reinforcement learning for electric vehicle fast charging station pricing game in electricity-transportation nexus. *IEEE Trans Ind Inform* 2024;20(4):6345–55.
- [99] Ye Yujian, Wang Hongru, Cui Tianxiang, Yang Xiaoying, Yang Shaofu, Zhang Min-Ling. Identifying generalizable equilibrium pricing strategies for charging service providers in coupled power and transportation networks. *Adv Appl Energy* 2023;12:100151.
- [100] Cui Tianxiang, Ding Shusheng, Jin Huan, Zhang Yongmin. Portfolio constructions in cryptocurrency market: A cvar-based deep reinforcement learning approach. *Econ Model* 2023;119:106078.
- [101] Vaswani Ashish, Shazeer Noam, Parmar Niki, Uszkoreit Jakob, Jones Llion, Gomez Aidan N, et al. Attention is all you need. In: *Advances in neural information processing systems*, vol. 30, 2017.
- [102] Yang Kun, Cheng Zishu, Li Mingchen, Wang Shouyang, Wei Yunjie. Fortify the investment performance of crude oil market by integrating sentiment analysis and an interval-based trading strategy. *Appl Energy* 2024;353:122102.
- [103] Gardner Jr. Everette S, Dannenbring David G. Forecasting with exponential smoothing: Some guidelines for model selection. *Decis Sci* 1980;11(2):370–83.
- [104] Hendershott Terrence, Livdan Dmitry, Schürhoff Norman. Are institutions informed about news? *J Financ Econ* 2015;117(2):249–87.